

Long–Short Portfolio Return Book

Cumulative return of each factor's long–short portfolio, net of 10 bp transaction cost.
270 factors over 133 months, ordered by in-sample L–S CAGR (last 48 months, net of cost).

HOW TO READ

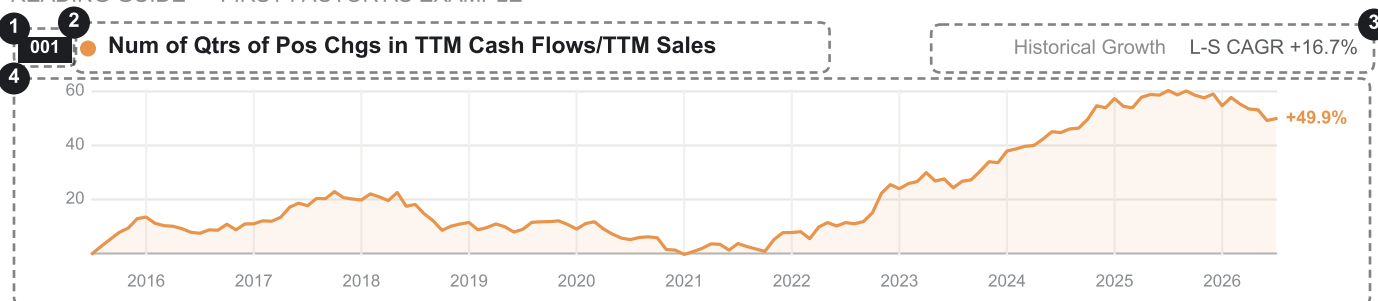
Each curve starts at 0 % in July 2015 and compounds monthly long–short returns. The line is coloured by the factor's style; the figure at the right edge is the period-end cumulative return.

FACTOR STYLES

Valuation	41	Price Momentum	56
Earnings Quality	55	Size	1
Analyst Expectations	24	Historical Growth	55
Capital Efficiency	28	Volatility	10

001 In current model portfolio (50 factors)

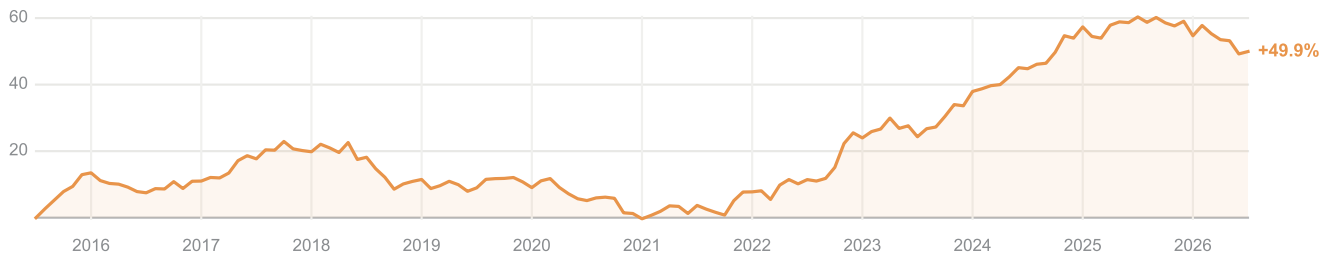
READING GUIDE — FIRST FACTOR AS EXAMPLE



001 ● Num of Qtrs of Pos Chgs in TTM Cash Flows/TTM Sales

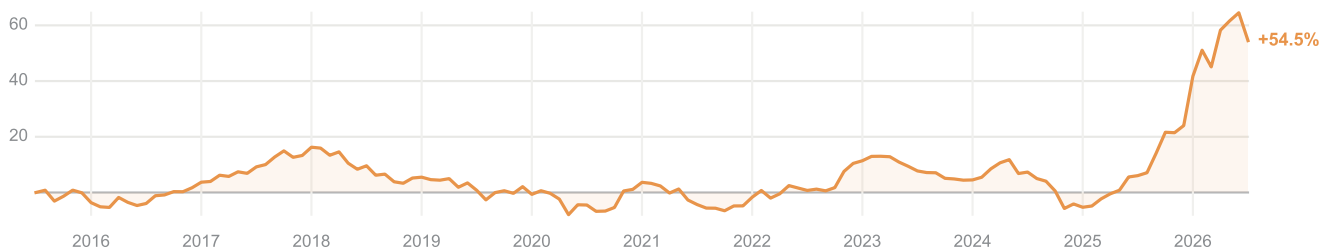
Historical Growth

CAGR +16.7%

**002 ● Trailing 12-Month Receivables and Inventories/Trailing 12-Month Sales**

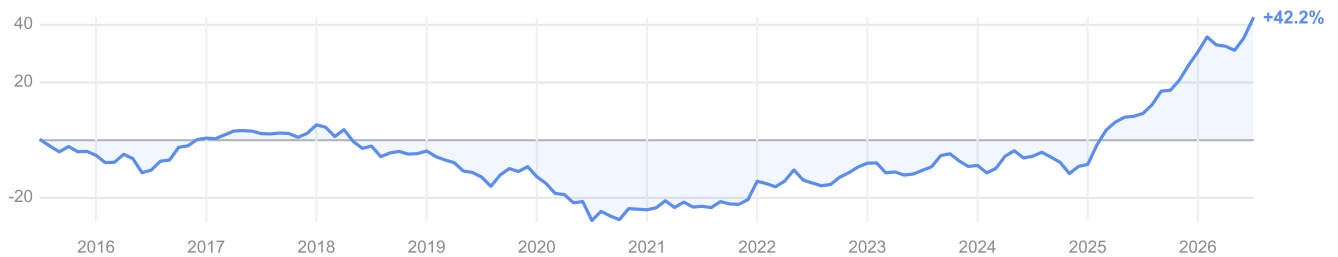
Historical Growth

CAGR +15.2%

**003 ● Pretax Income Per Share to Price**

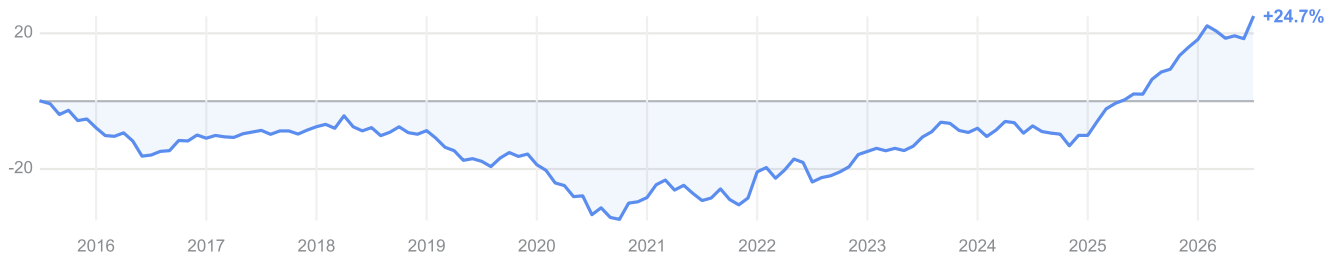
Valuation

CAGR +12.4%

**004 ● Current Liability to Price**

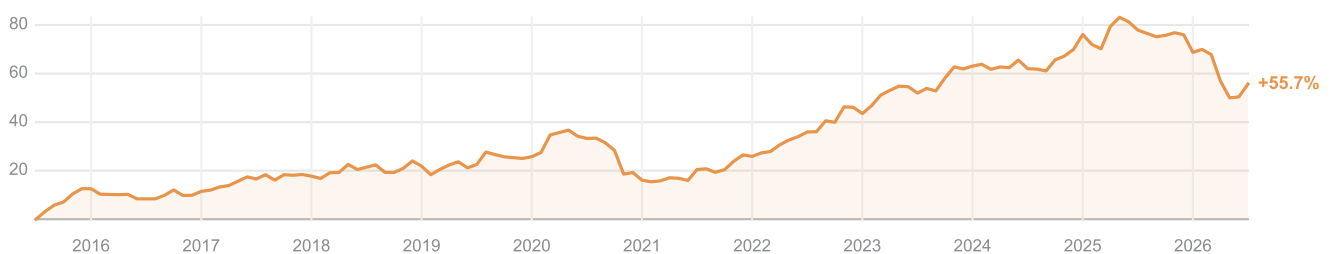
Valuation

CAGR +11.9%

**005 ● Number of Consecutive Qtrs of Pos. Changes in TTM Cash Flows**

Historical Growth

CAGR +11.9%

**006 ● Operating Earnings to Assets Ratio**

Earnings Quality

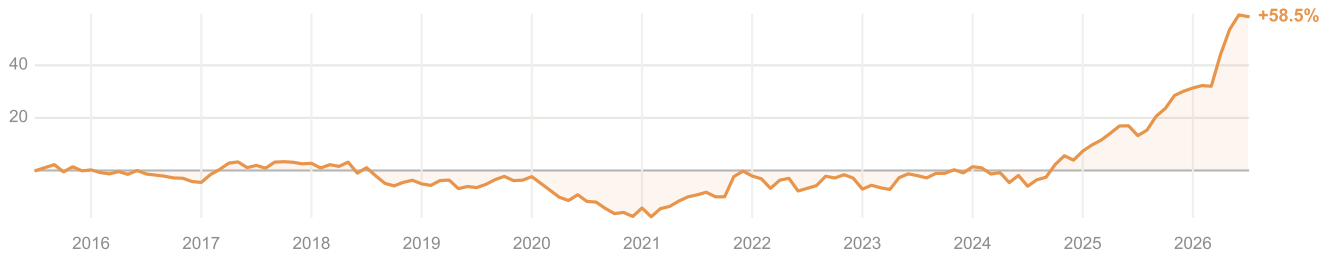
CAGR +11.8%



007 ● Change of Asset Turnover

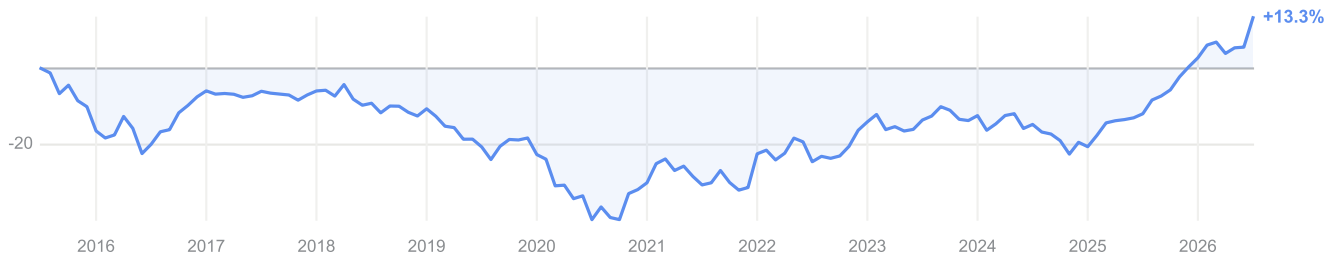
Historical Growth

CAGR +11.4%

**008 ● Assets to Price Ratio**

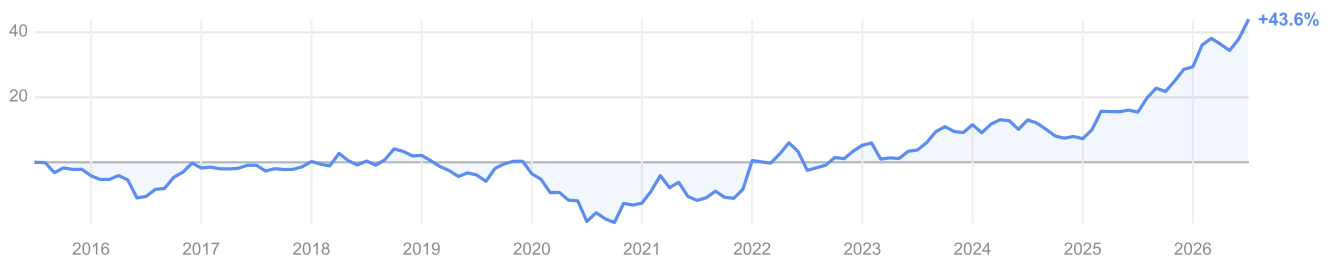
Valuation

CAGR +11.3%

**009 ● Sales to Price Ratio**

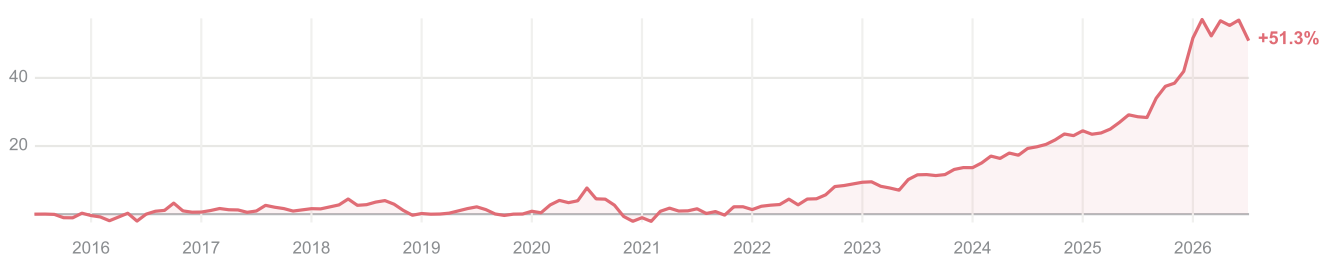
Valuation

CAGR +10.6%

**010 ● Unexpected Inventory Change**

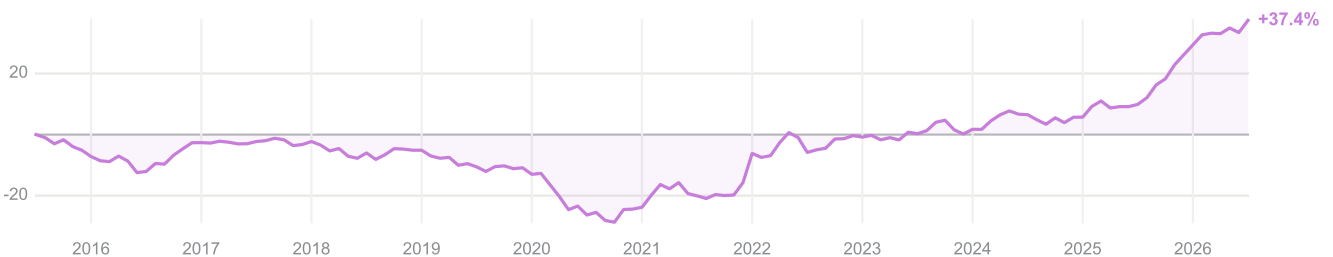
Earnings Quality

CAGR +10.5%

**011 ● Log of Total Last Quarter Assets**

Size

CAGR +10.3%

**012 ● Inverse P/E Ratio Adj for Growth and Yield**

Valuation

CAGR +10.3%



013 ● Indicated Annual Dividends to Price Ratio

Valuation CAGR +10.1%



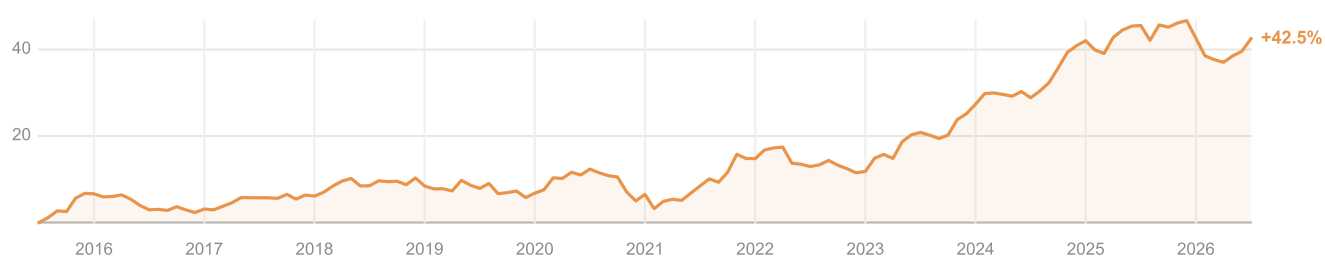
014 ● Cash to Price

Valuation CAGR +10.0%



015 ● Adj 1Y Chg in Asset Adjusted FCF

Historical Growth CAGR +9.8%



016 ● Long Term Debt to Equity Ratio

Capital Efficiency CAGR +9.8%



017 ● Return on Assets

Capital Efficiency CAGR +9.8%



018 ● Number of Consecutive Qtrs of Positive Changes in TTM Quarterly Earnings

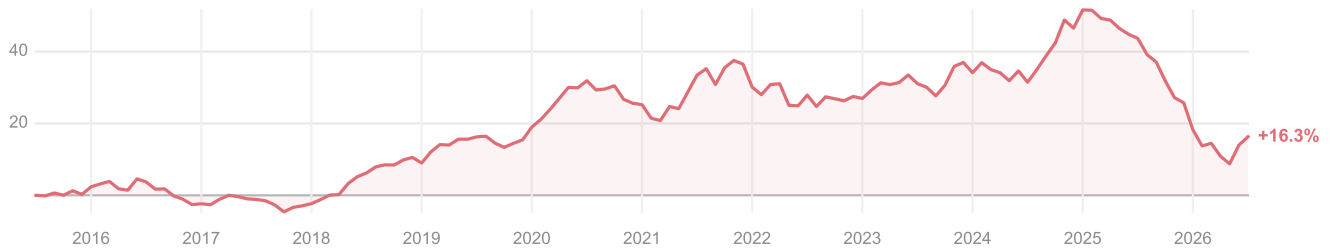
Historical Growth CAGR +9.7%



019 ● Gross Profit Margin

Earnings Quality

CAGR +9.7%

**020 ● Adjusted 3M Revision in FY1 EPS Estimate**

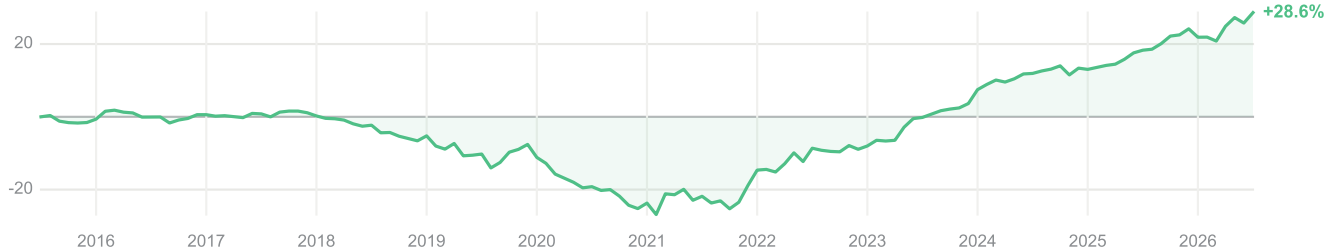
Analyst Expectations

CAGR +9.7%

**021 ● Adjusted Inverse PEG**

Analyst Expectations

CAGR +9.7%

**022 ● Analyst Dispersion for FY2 EPS**

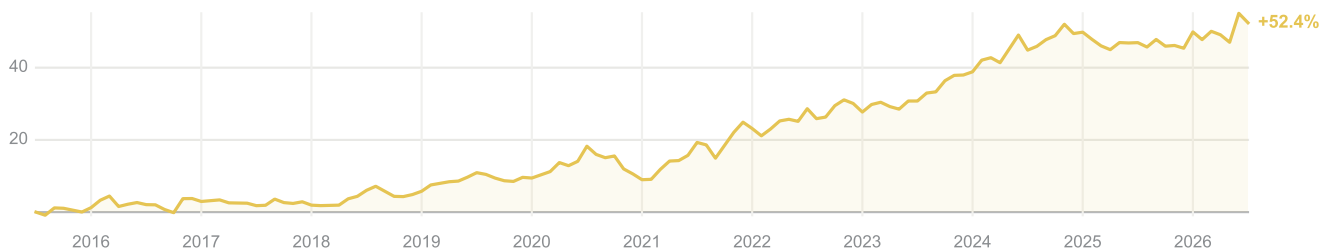
Analyst Expectations

CAGR +9.6%

**023 ● Return on Equity**

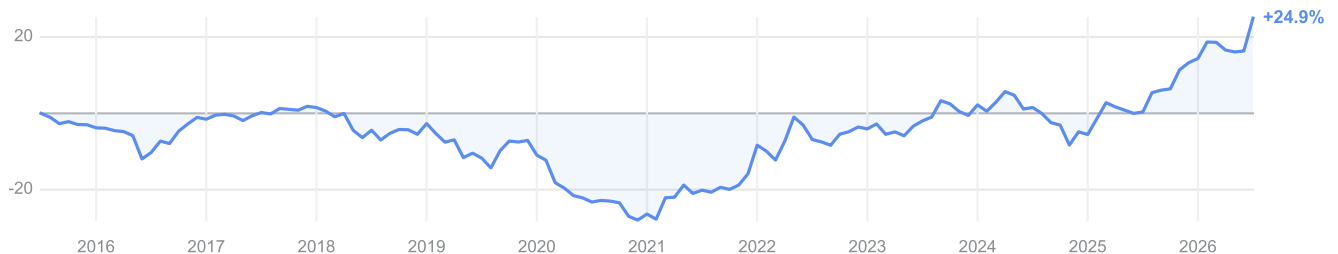
Capital Efficiency

CAGR +9.5%

**024 ● Forward Earnings to Price**

Valuation

CAGR +9.5%



025 ● **Sharpe Ratio**

Price Momentum

CAGR +9.4%

026 ● **12M Realized Price Volatility**

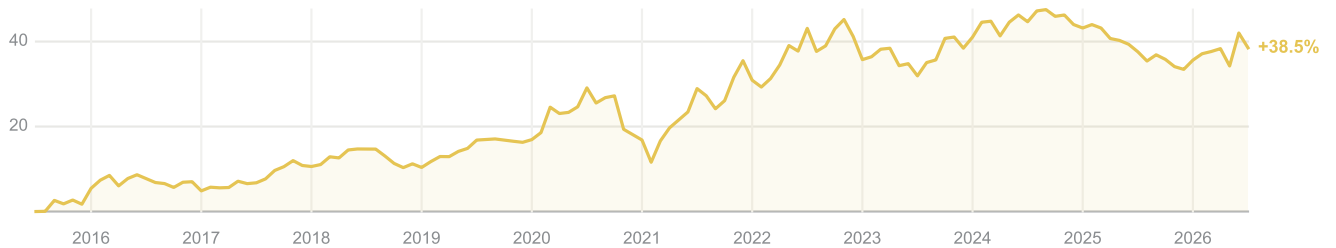
Volatility

CAGR +9.4%

027 ● **Return on Invested Capital**

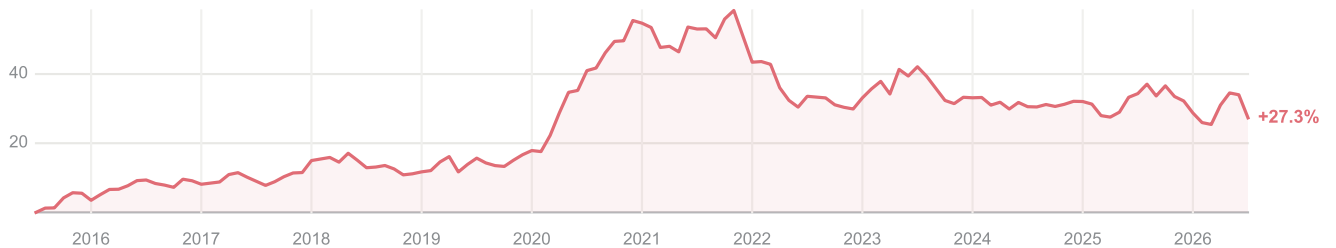
Capital Efficiency

CAGR +9.4%

028 ● **Quick Ratio**

Earnings Quality

CAGR +9.3%

029 ● **3Y Chg in Operating Cash Flow to Price**

Historical Growth

CAGR +9.2%

030 ● **Operating Earnings to Price Ratio**

Valuation

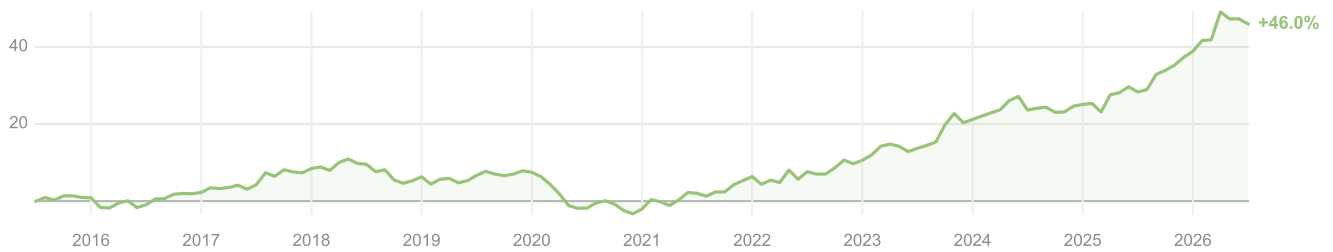
CAGR +9.1%



031 ● Short Interest Trading Volume

Price Momentum

CAGR +9.0%

**032 ● Trailing 12-Month Earnings/Trailing 12-Month Sales**

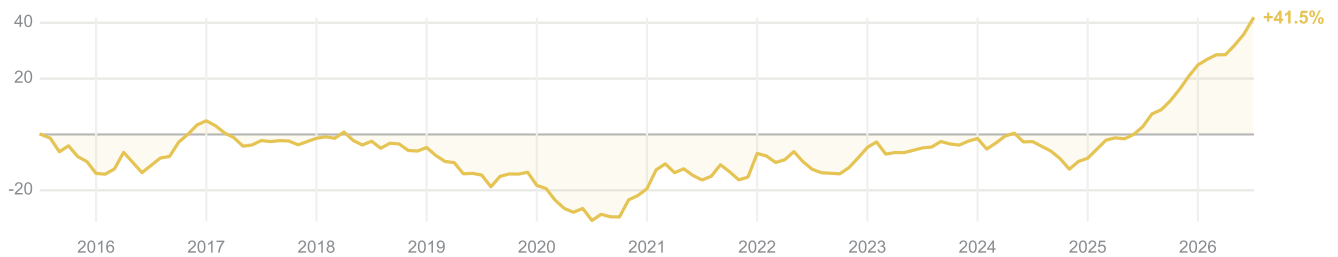
Earnings Quality

CAGR +9.0%

**033 ● Tobin's Q Ratio**

Capital Efficiency

CAGR +8.9%

**034 ● ROA 20 Qtr Standard Deviation**

Earnings Quality

CAGR +8.8%

**035 ● Adj Growth Flow to Price**

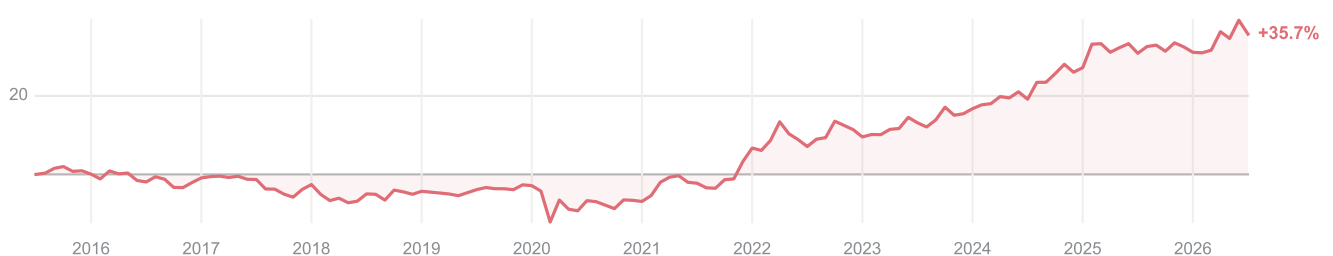
Valuation

CAGR +8.7%

**036 ● Working Capital to Sales**

Earnings Quality

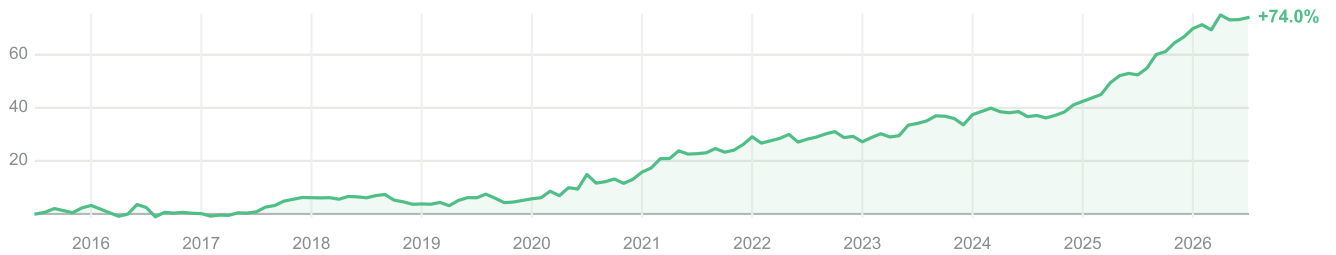
CAGR +8.4%



037 ● 3M Revision in FY1 EPS Estimate

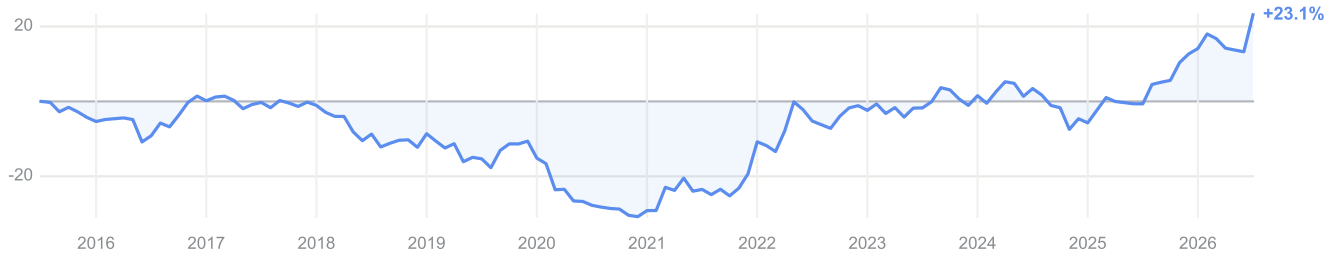
Analyst Expectations

CAGR +8.3%

**038 ● Adjusted Forward Earnings to Price**

Valuation

CAGR +8.2%

**039 ● Normalized EP**

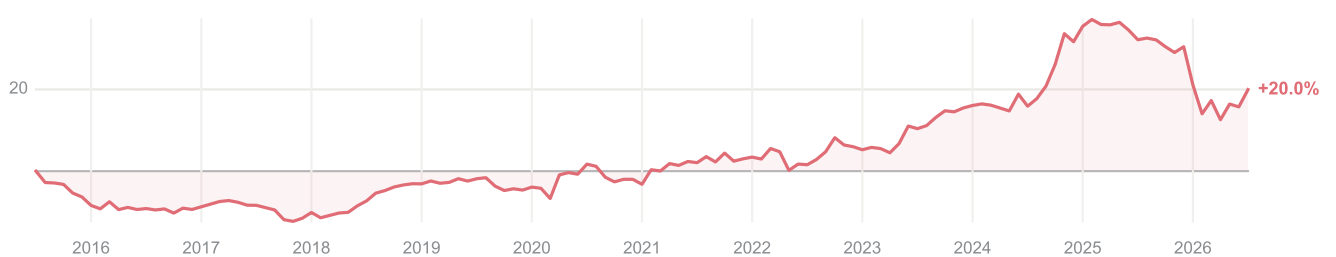
Valuation

CAGR +8.2%

**040 ● Balance Sheet Accruals**

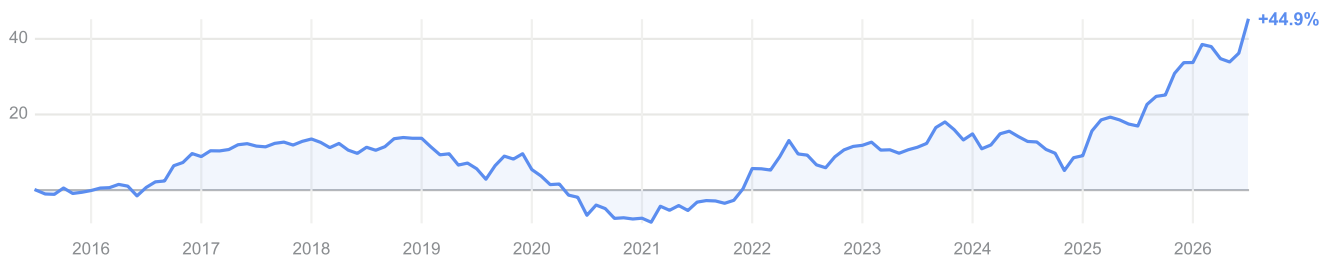
Earnings Quality

CAGR +8.1%

**041 ● Earnings to Price**

Valuation

CAGR +8.1%

**042 ● Slope of 52W Price Trend**

Price Momentum

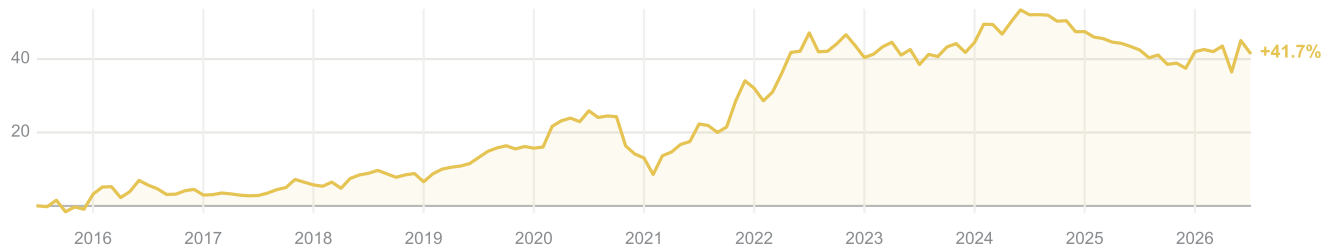
CAGR +8.1%



043 ● EBITDA to Assets

Capital Efficiency

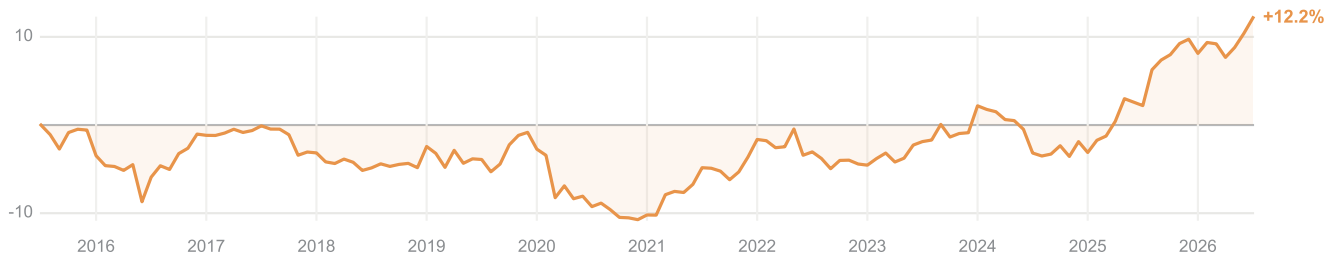
CAGR +8.0%



044 ● 3Y Chg in Cash Flow to Price

Historical Growth

CAGR +8.0%



045 ● Consecutive Quarters of Declines in TTM Overhead/Sales

Historical Growth

CAGR +8.0%



046 ● Cash Flow to Total Asset

Earnings Quality

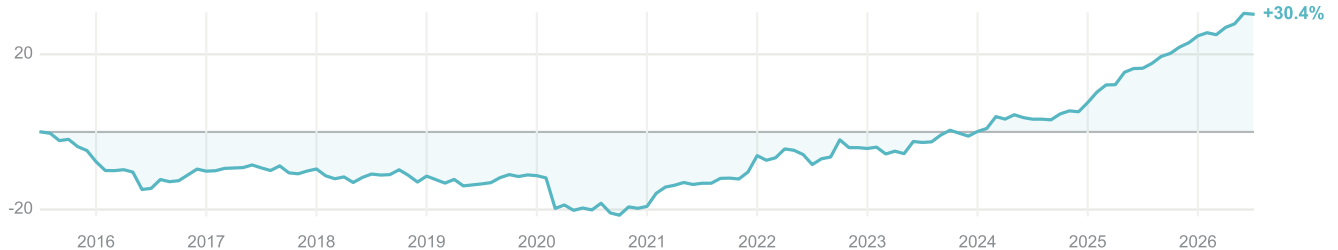
CAGR +8.0%



047 ● Book Leverage

Volatility

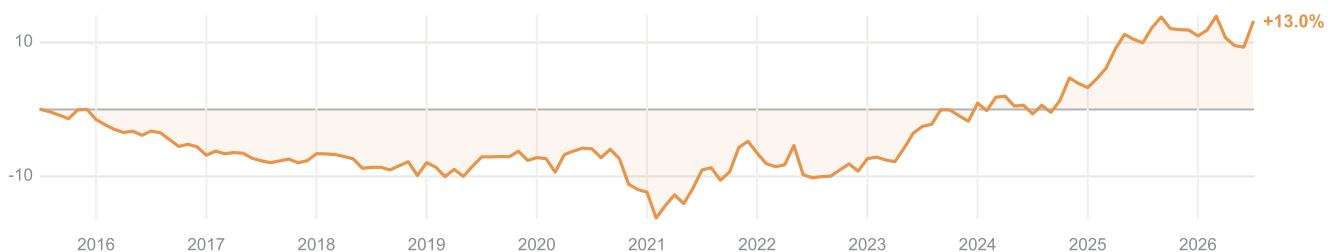
CAGR +7.9%



048 ● 1Y Chg in Sales to Price

Historical Growth

CAGR +7.8%



049 ● 60M CAPM Beta

Volatility

CAGR +7.7%



050 ● Adj Free Cash Flow to Price

Valuation

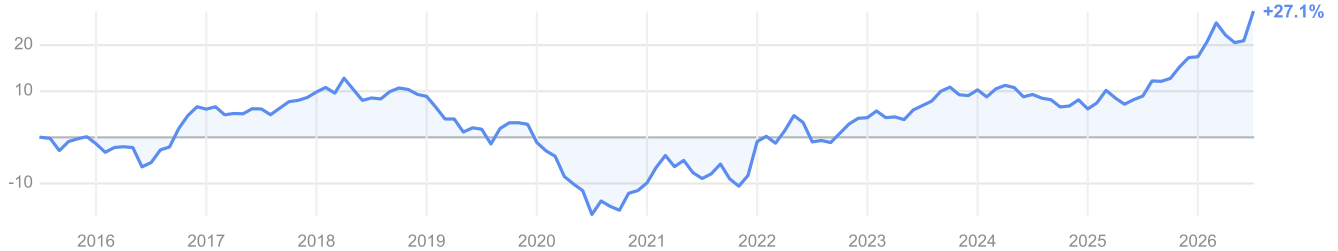
CAGR +7.6%



051 ● Sales to EV Ratio

Valuation

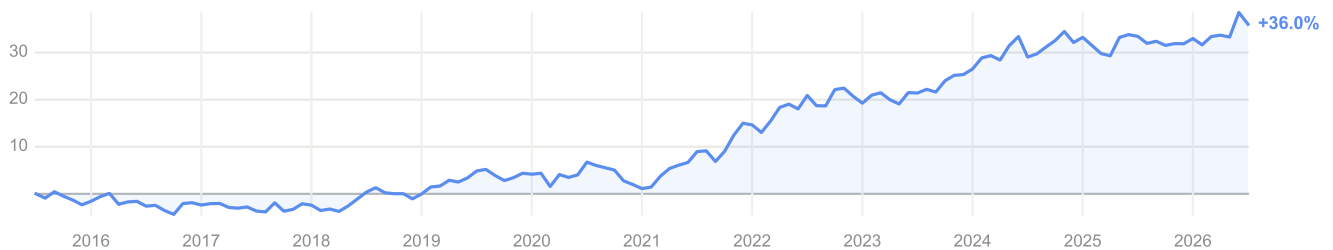
CAGR +7.6%



052 ● Cash Flow to Equity

Valuation

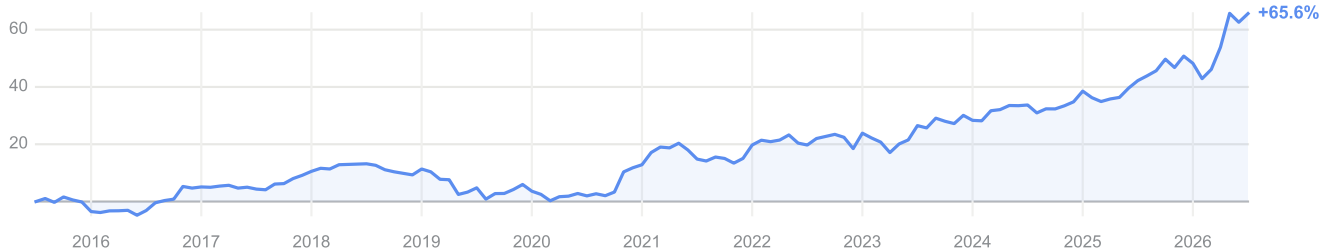
CAGR +7.6%



053 ● Cash to Adjusted Enterprise Value

Valuation

CAGR +7.6%

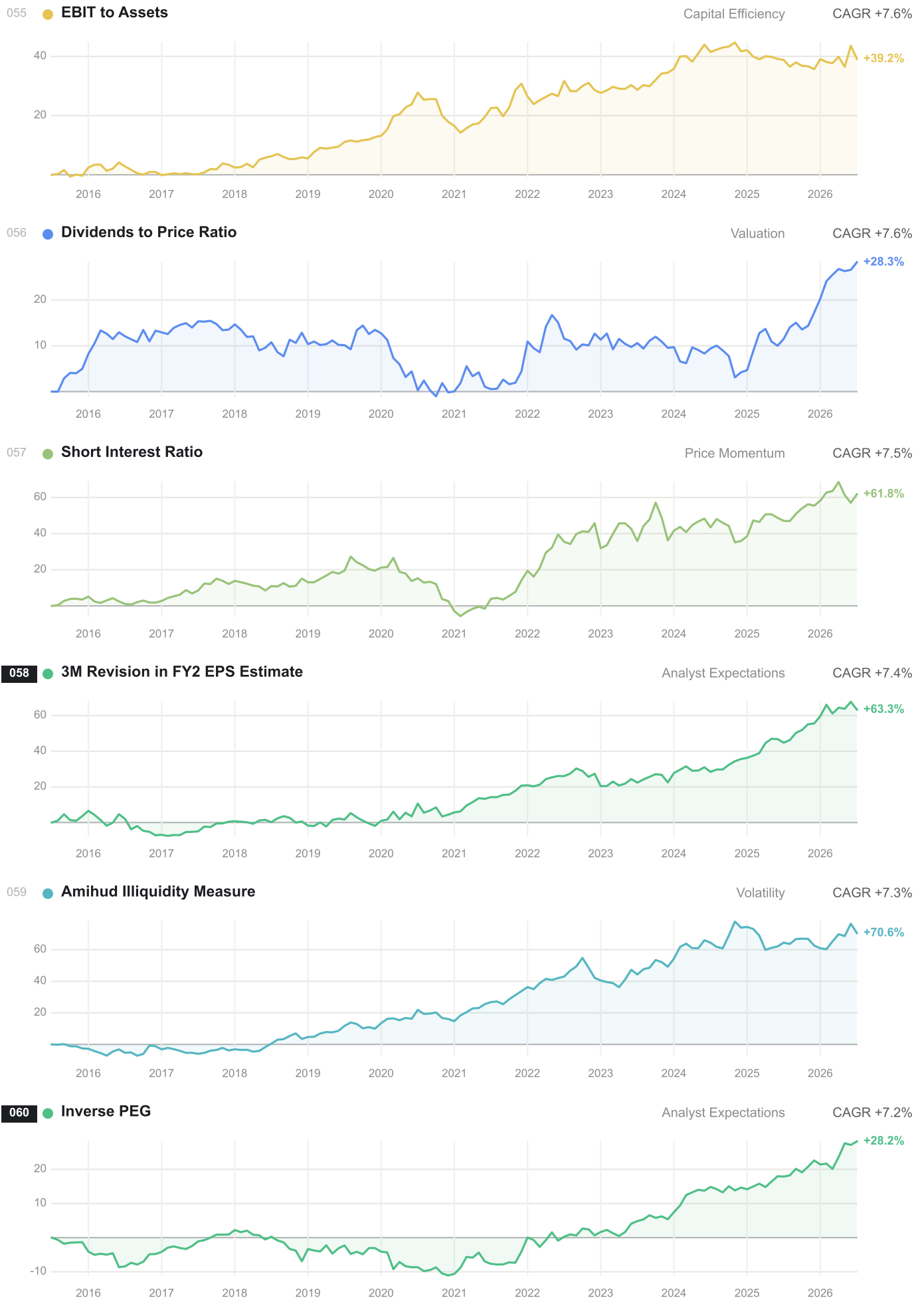


054 ● Expected LTG

Analyst Expectations

CAGR +7.6%





061 ● Adjusted EBIT to Price

Valuation

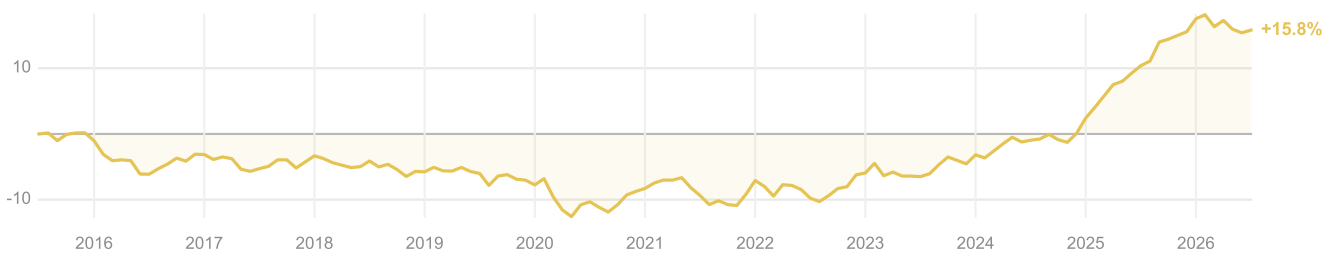
CAGR +7.1%



062 ● Capital Expenditure to Total Assets

Capital Efficiency

CAGR +7.1%



063 ● Net Cash Flow to Price

Valuation

CAGR +7.1%



064 ● 3Y Chg in Operating Cash Flow per Share

Historical Growth

CAGR +7.1%



065 ● 1Y Chg in Operating Margin

Historical Growth

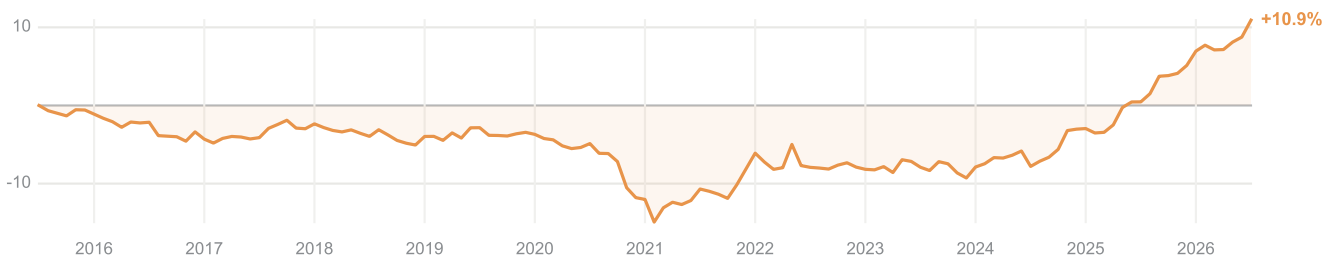
CAGR +7.1%



066 ● 1Y Chg in Cash Flow to Price

Historical Growth

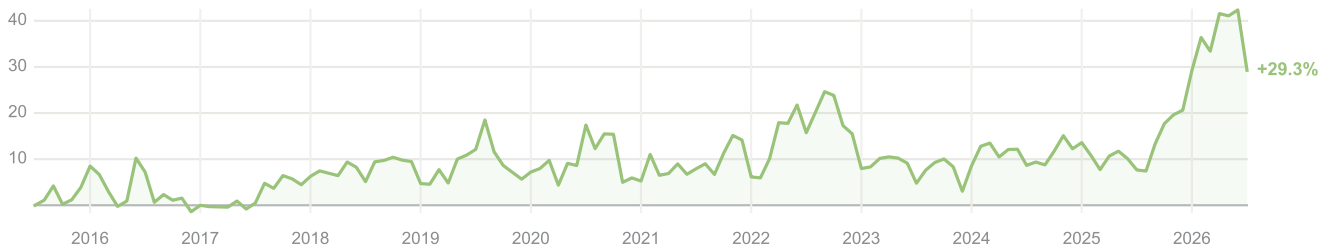
CAGR +7.1%



067 ● 50 Day to 200 Day Stock Price

Price Momentum

CAGR +7.1%



068 ● Adjusted Book to Enterprise Value

Valuation

CAGR +7.0%



069 ● 1Y Chg in Asset Adjusted EPS

Historical Growth

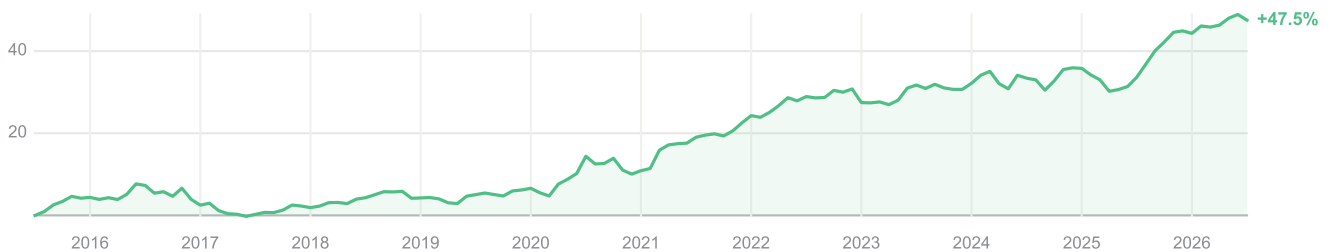
CAGR +7.0%



070 ● Earnings Surprise

Analyst Expectations

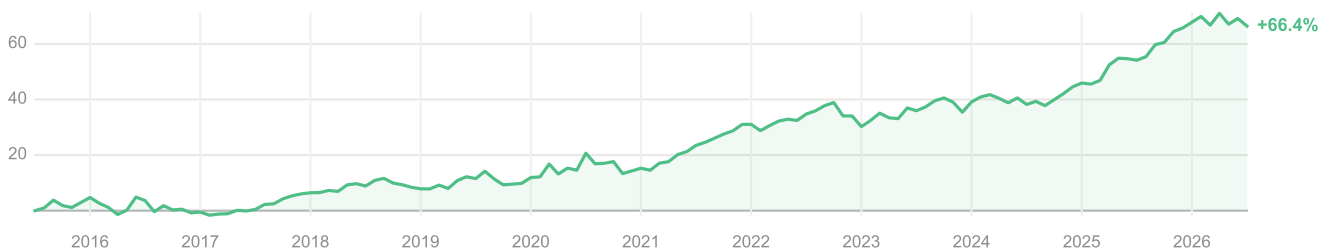
CAGR +7.0%



071 ● Street Revision Magnitude

Analyst Expectations

CAGR +7.0%



072 ● Cash Flow to Invested Capital

Earnings Quality

CAGR +7.0%



073 ● Sales to Invested Capital

Earnings Quality

CAGR +7.0%

**074 ● 39W Lagged Return**

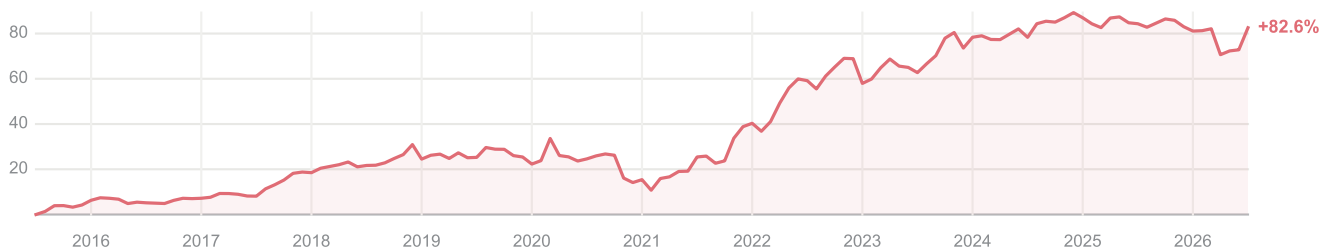
Price Momentum

CAGR +7.0%

**075 ● Operating Cash Flow to Invested Capital**

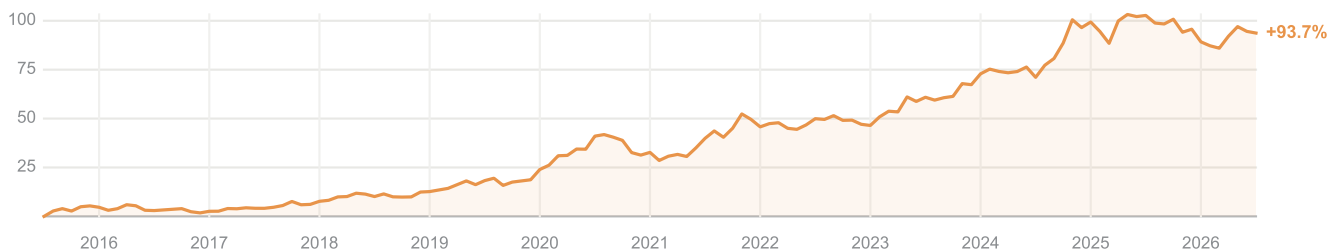
Earnings Quality

CAGR +7.0%

**076 ● Adjusted 3Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth

CAGR +7.0%

**077 ● 1Y Chg in Earnings to Price**

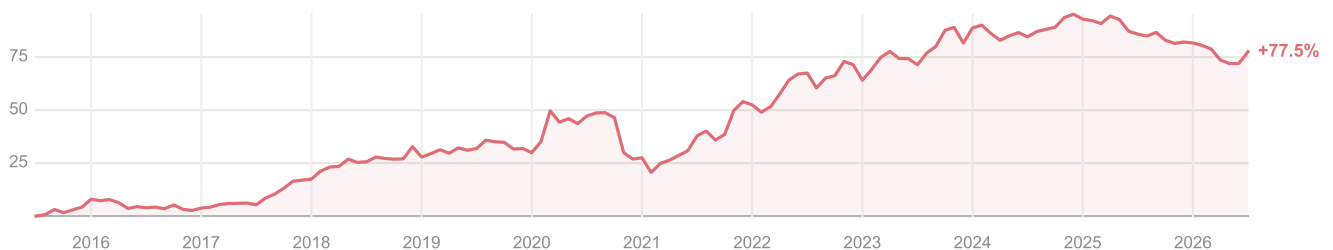
Historical Growth

CAGR +7.0%

**078 ● Operating Cash Flow to Asset**

Earnings Quality

CAGR +6.9%



079 ● **EBITDA to Enterprise Value**

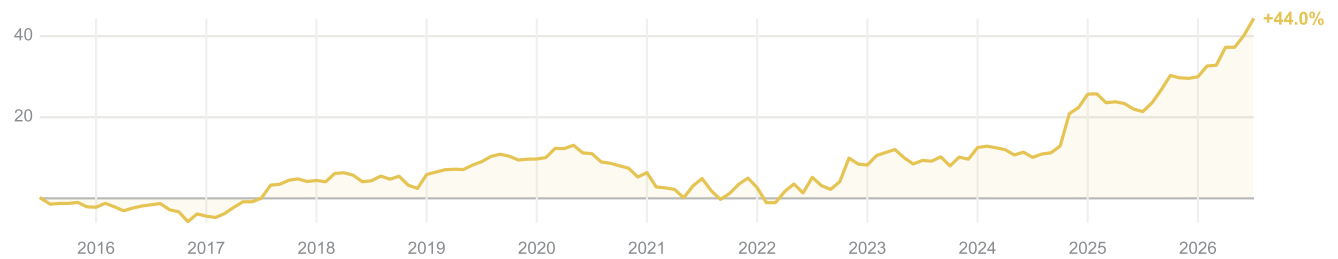
Valuation

CAGR +6.8%

080 ● **1Y Chg in Stock Buybacks**

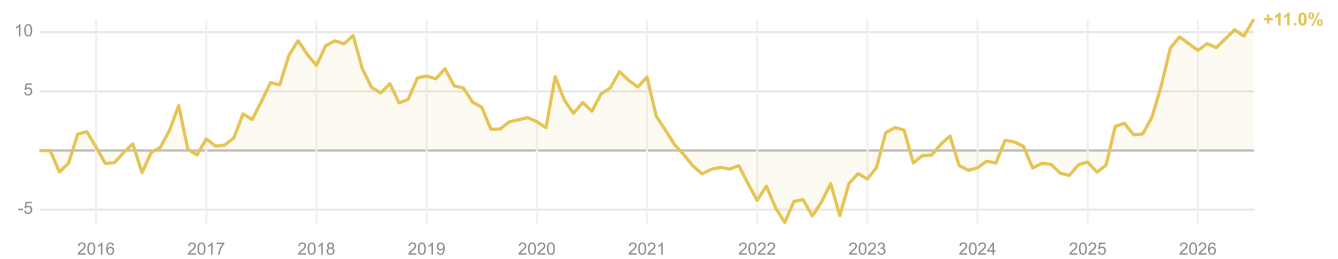
Capital Efficiency

CAGR +6.8%

081 ● **Long Term Debt to Assets Ratio**

Capital Efficiency

CAGR +6.8%

082 ● **Institution Ownership Level - Active Manager**

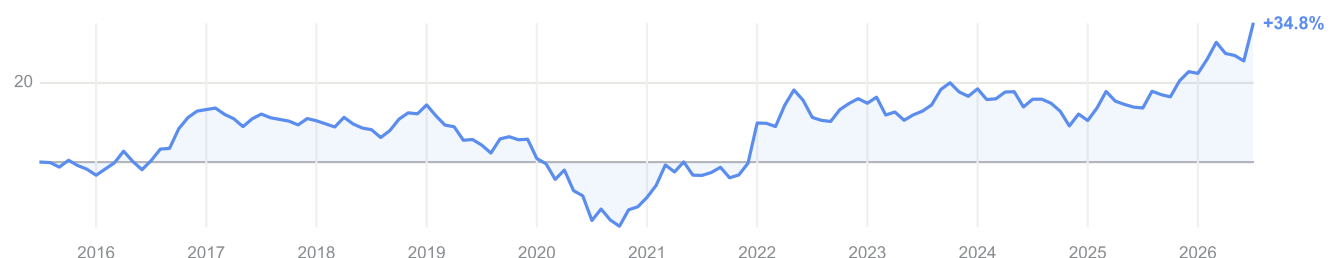
Price Momentum

CAGR +6.8%

083 ● **Operating Cash Flow to Price**

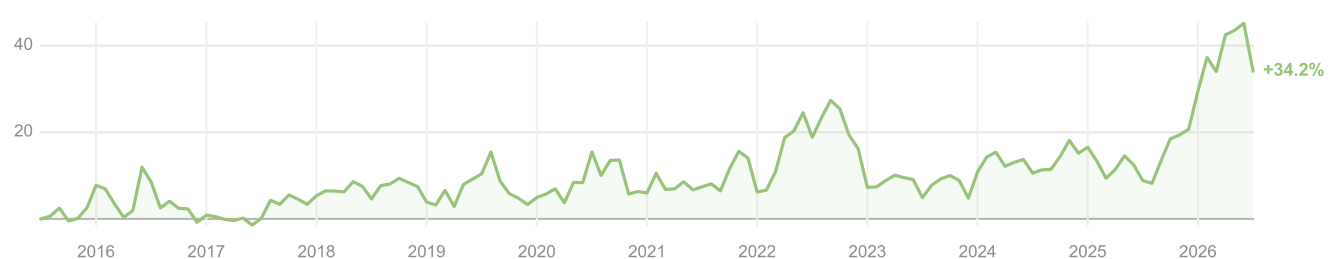
Valuation

CAGR +6.8%

084 ● **15/36 Week Price Ratio**

Price Momentum

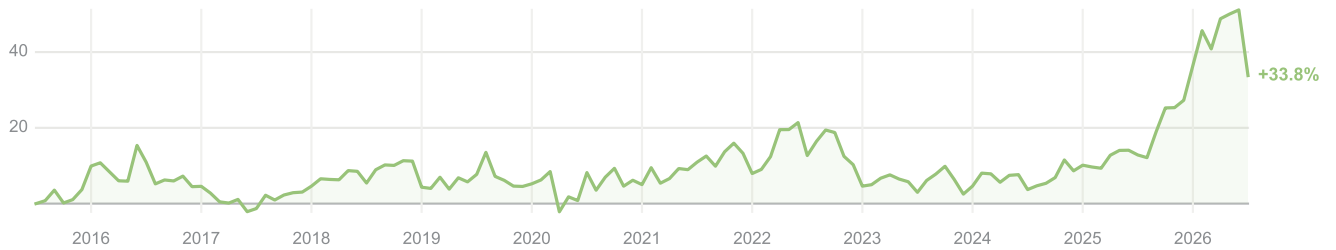
CAGR +6.8%



085 ● **26W Relative Price Strength**

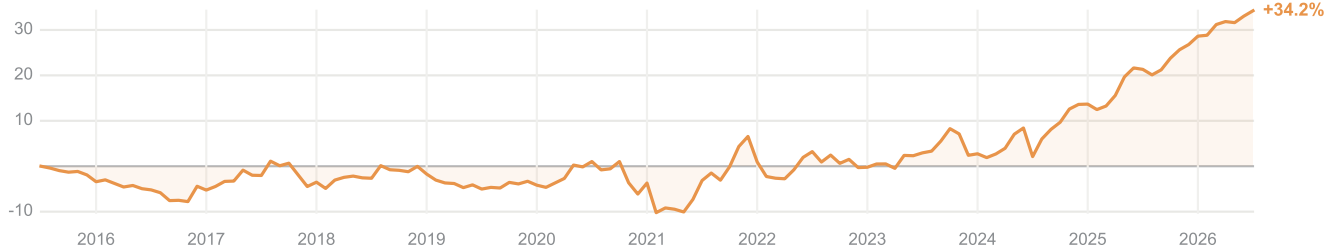
Price Momentum

CAGR +6.8%

086 ● **Adj 1Y Chg in Earnings per Share**

Historical Growth

CAGR +6.7%

087 ● **90 Day Coefficient of Variation**

Volatility

CAGR +6.7%

088 ● **Solvency Ratio**

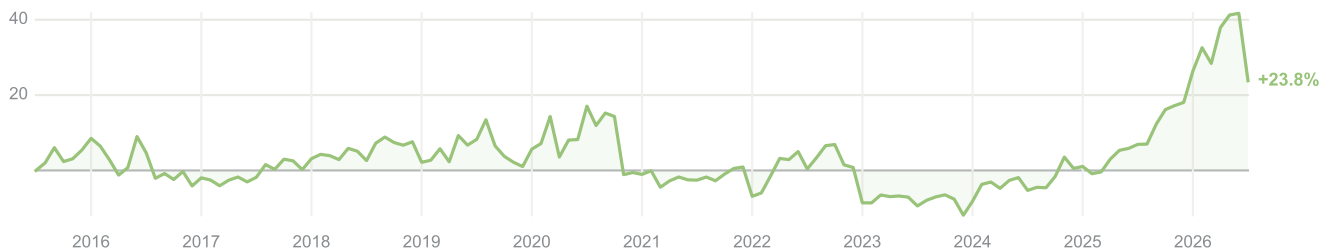
Valuation

CAGR +6.7%

089 ● **9 Month Price Momentum**

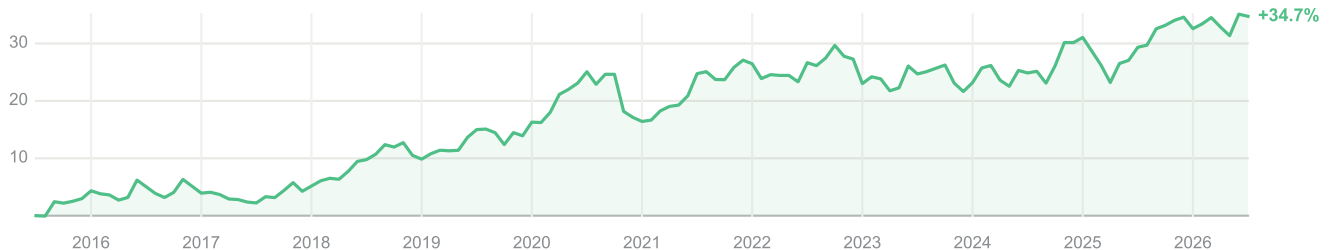
Price Momentum

CAGR +6.6%

090 ● **Standardized Unexpected Earnings**

Analyst Expectations

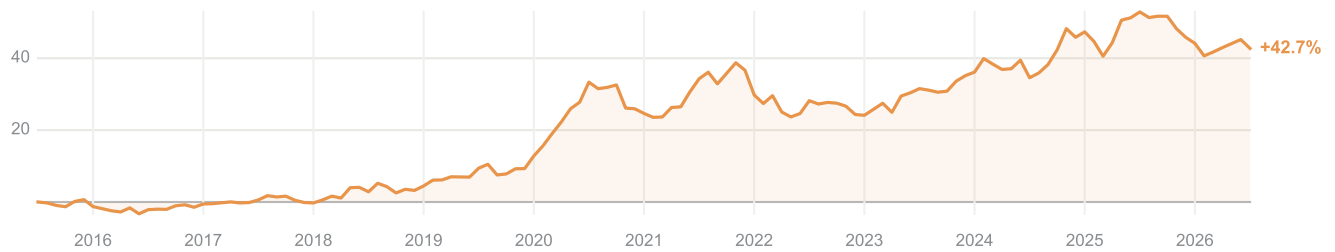
CAGR +6.6%



091 ● 3Y Chg in Asset Adjusted Operating Cash Flow

Historical Growth

CAGR +6.6%



092 ● Adj. EBITDA to Enterprise Value

Valuation

CAGR +6.6%



093 ● Operating Cash Flow to Enterprise Value

Valuation

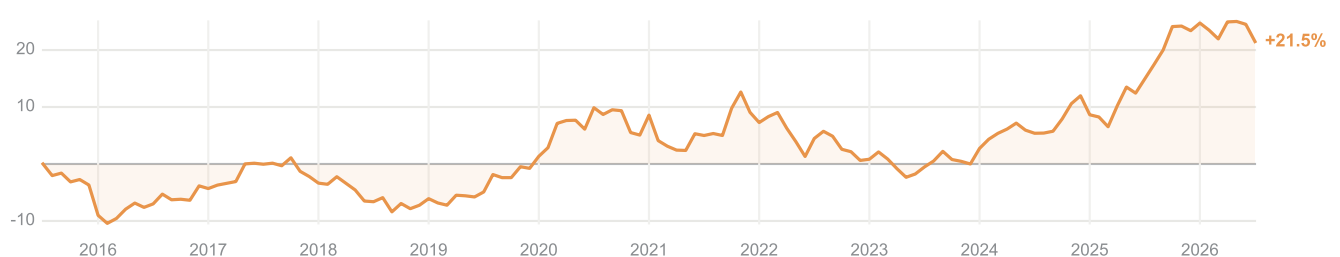
CAGR +6.5%



094 ● 12-Quarter Trendline in Trailing 12-Month Earnings

Historical Growth

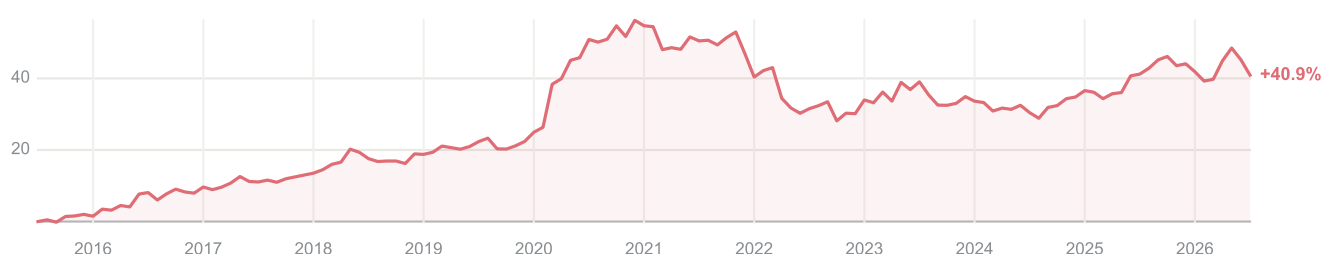
CAGR +6.5%



095 ● Cash Ratio

Earnings Quality

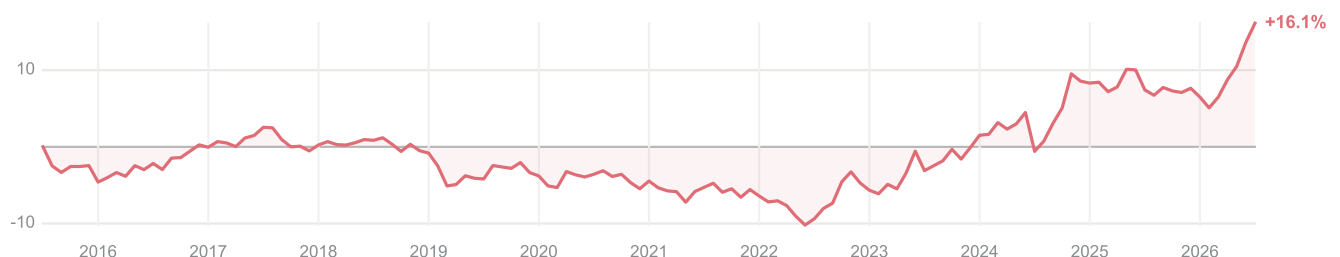
CAGR +6.5%



096 ● Inventory Turnover

Earnings Quality

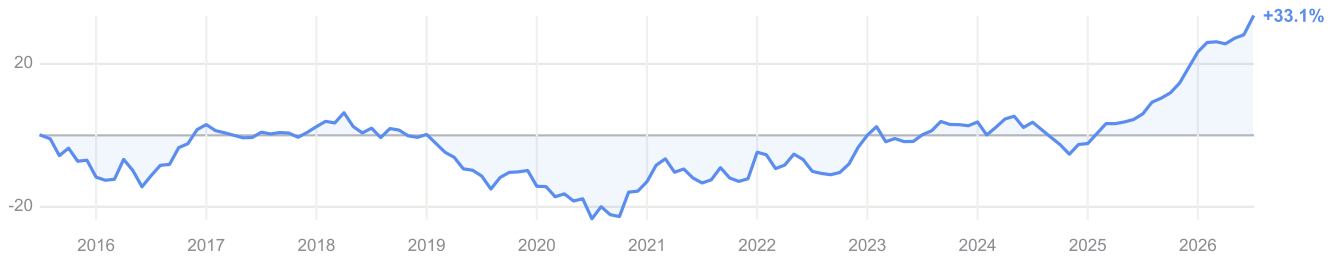
CAGR +6.4%



097 ● **Book to Price**

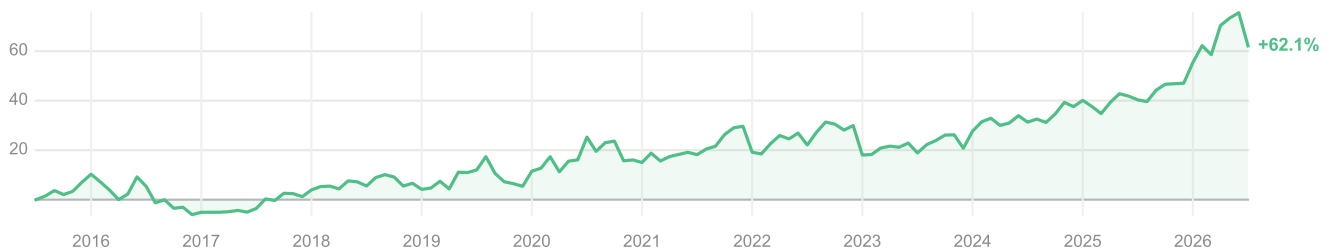
Valuation

CAGR +6.4%

098 ● **6M Chg in Target Price**

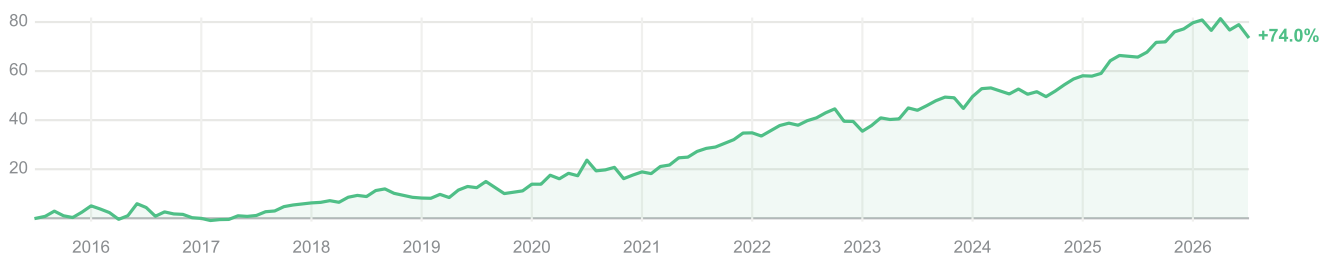
Analyst Expectations

CAGR +6.4%

099 ● **Adjusted Revision Magnitude**

Analyst Expectations

CAGR +6.4%

100 ● **Expected Earnings Growth: Fiscal Year 2/Fiscal Year 1**

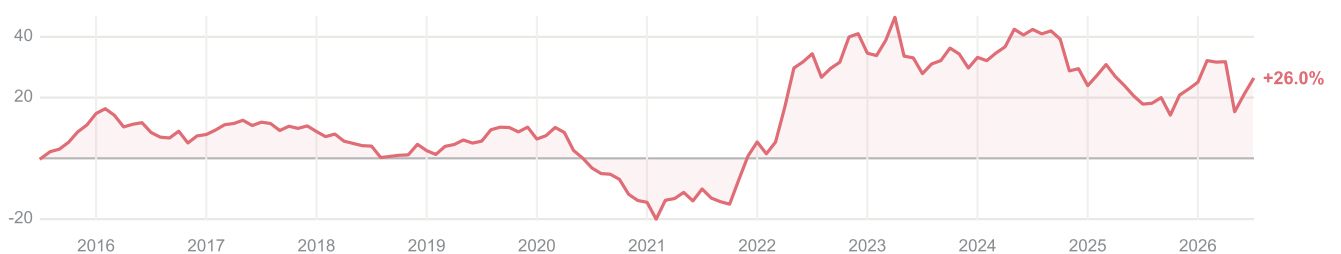
Historical Growth

CAGR +6.4%

101 ● **EBITDA Margin**

Earnings Quality

CAGR +6.3%

102 ● **Capital Expenditure to Sales**

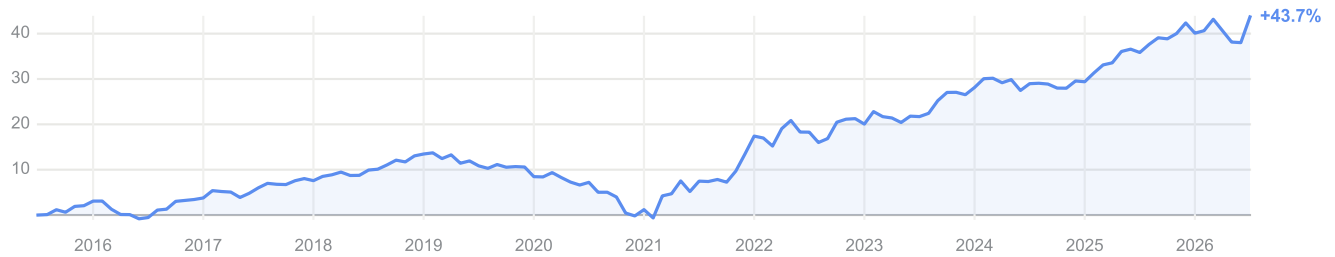
Earnings Quality

CAGR +6.3%



103 ● Free Cash Flow to Enterprise Value

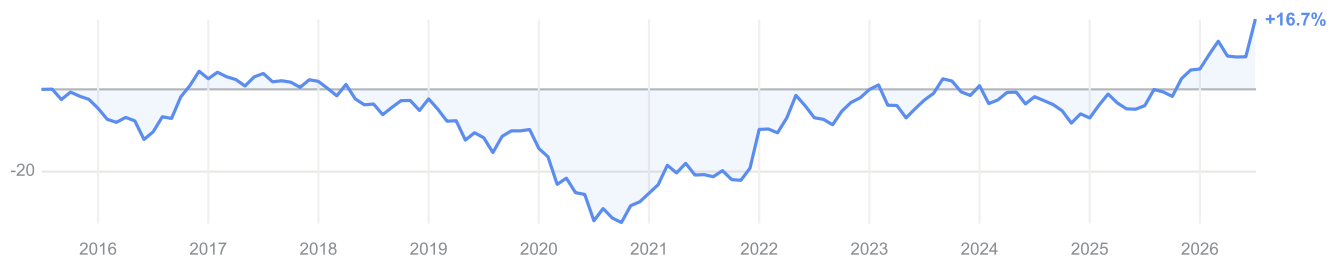
Valuation CAGR +6.3%

**104 ● 1M Realized Price Volatility**

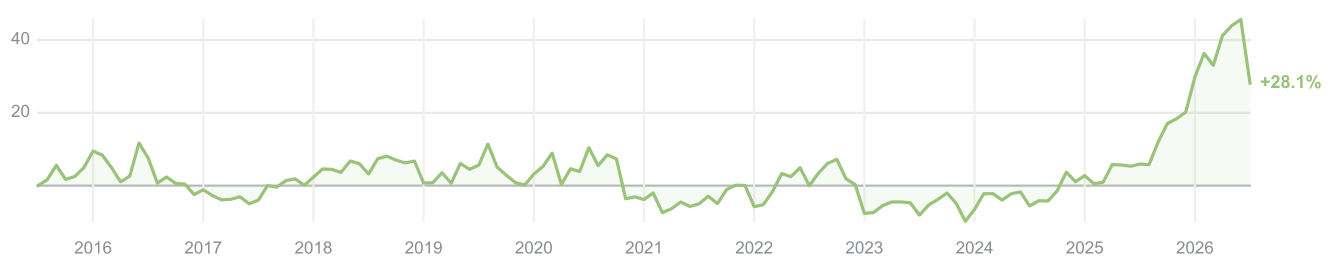
Volatility CAGR +6.3%

**105 ● EBITDA to Price**

Valuation CAGR +6.3%

**106 ● 4W to 52W Price Oscillator**

Price Momentum CAGR +6.3%

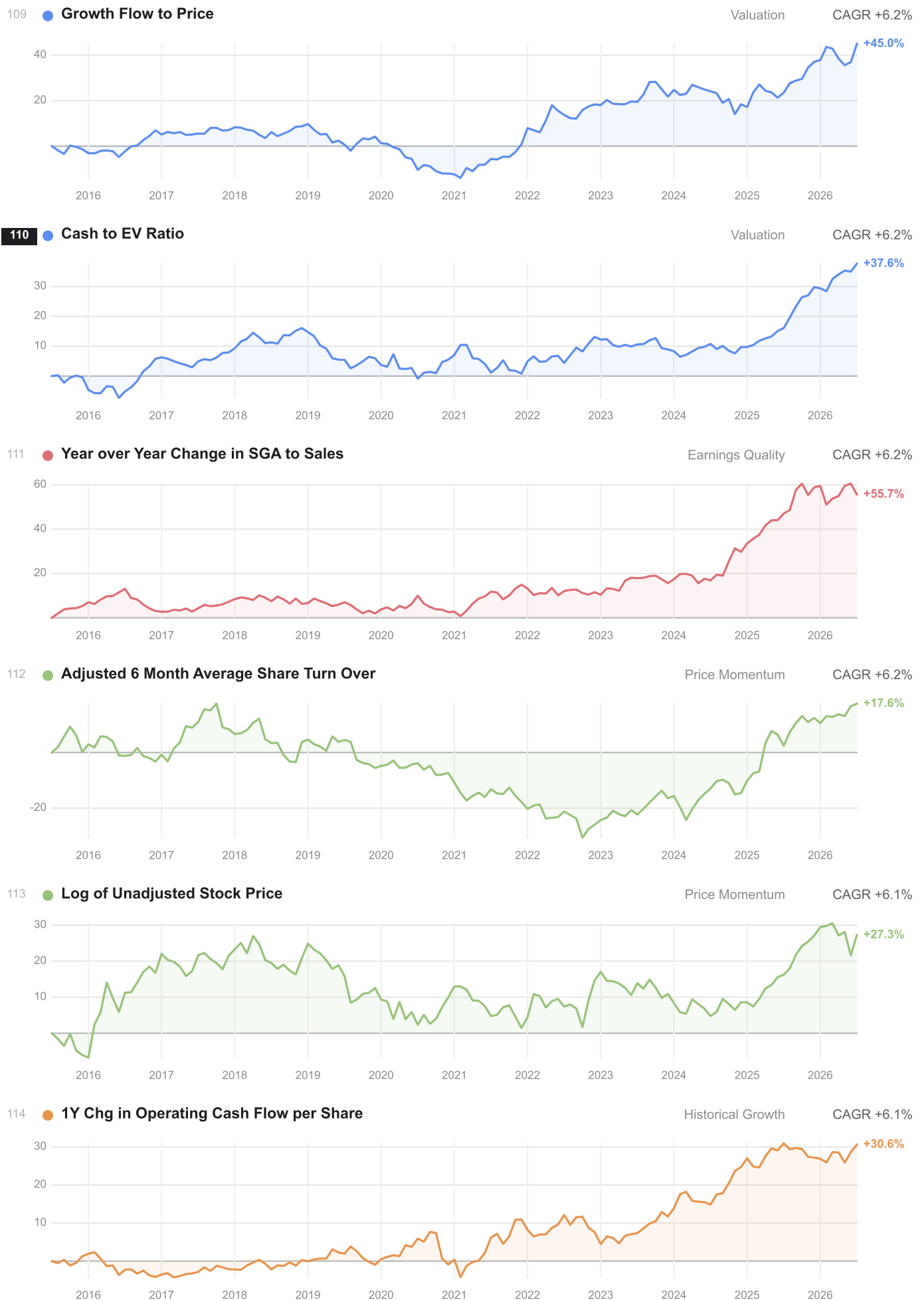
**107 ● Percent Above 260-Day Low**

Price Momentum CAGR +6.3%

**108 ● 1Y Chg in Asset Adjusted Cash Flow**

Historical Growth CAGR +6.2%





115 ● **Analyst Dispersion for FY1 EPS**

Analyst Expectations

CAGR +6.1%

116 ● **Cash Flow Return on Invested Capital**

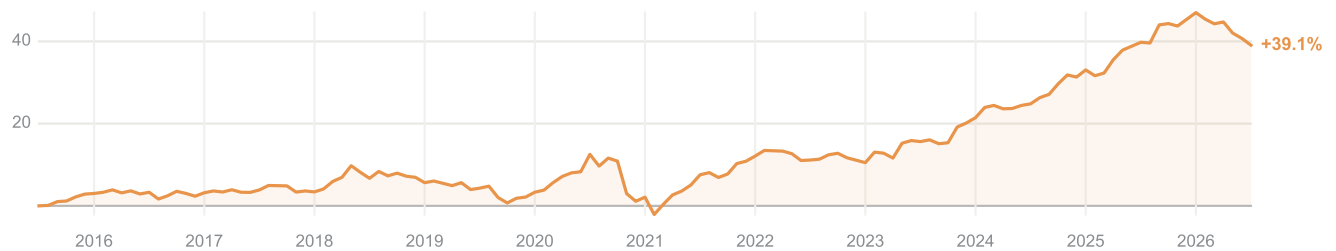
Capital Efficiency

CAGR +6.1%

117 ● **1Y Chg in Asset Adjusted Free Cash Flow**

Historical Growth

CAGR +6.0%

118 ● **Stability of Change in Number of Holders**

Price Momentum

CAGR +6.0%

119 ● **Operating Cash Flow Ratio**

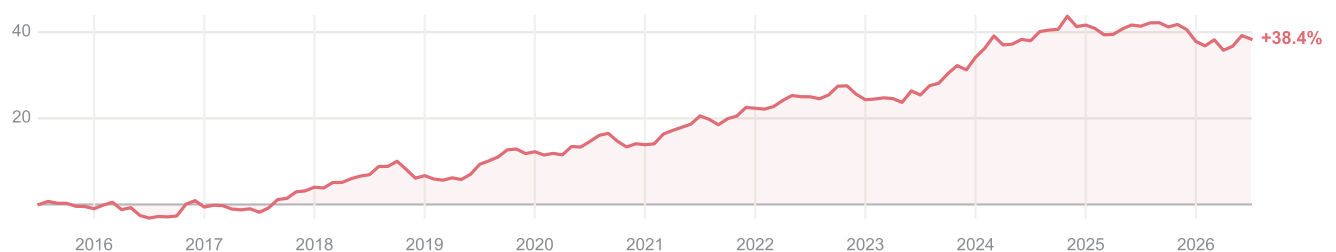
Capital Efficiency

CAGR +6.0%

120 ● **Equity Turnover**

Earnings Quality

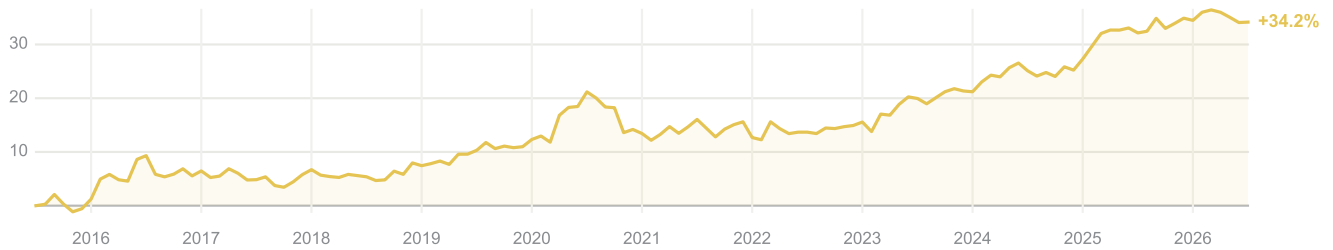
CAGR +6.0%



121 ● Receivables Turnover Ratio

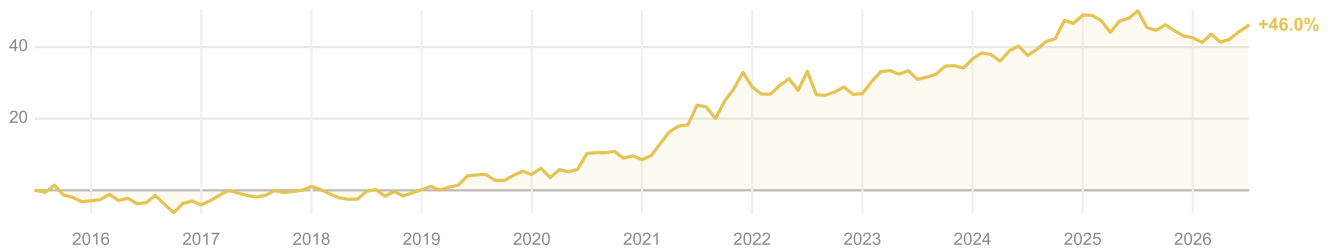
Capital Efficiency

CAGR +6.0%

**122 ● Adjusted Interest Coverage Ratio**

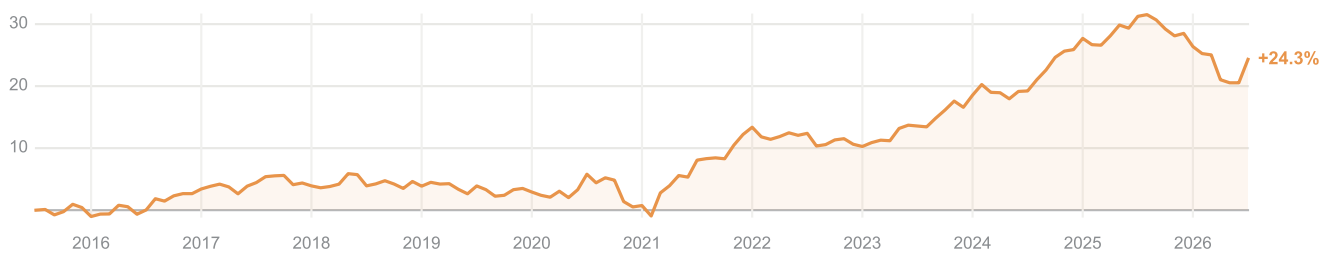
Capital Efficiency

CAGR +5.9%

**123 ● 3Y Chg in Free Cash Flow to Price**

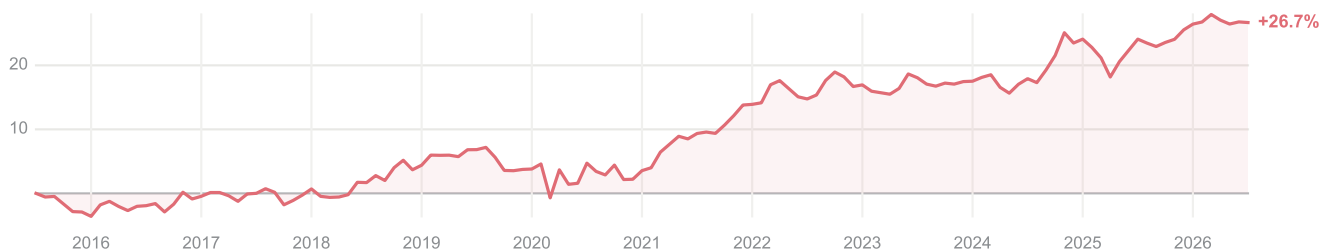
Historical Growth

CAGR +5.9%

**124 ● Working Capital Accruals**

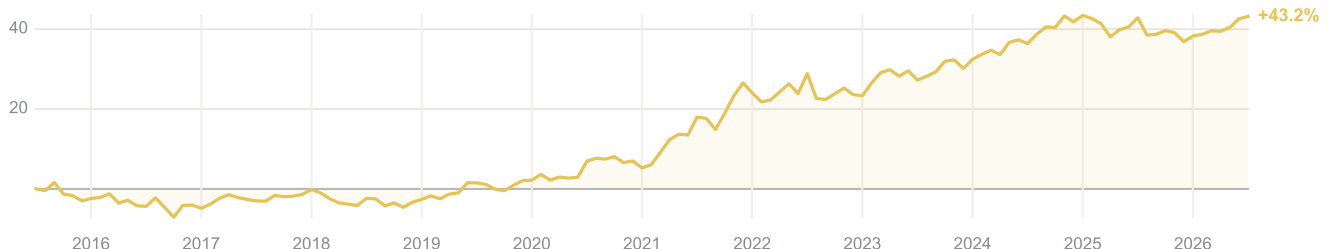
Earnings Quality

CAGR +5.8%

**125 ● Interest Coverage Ratio**

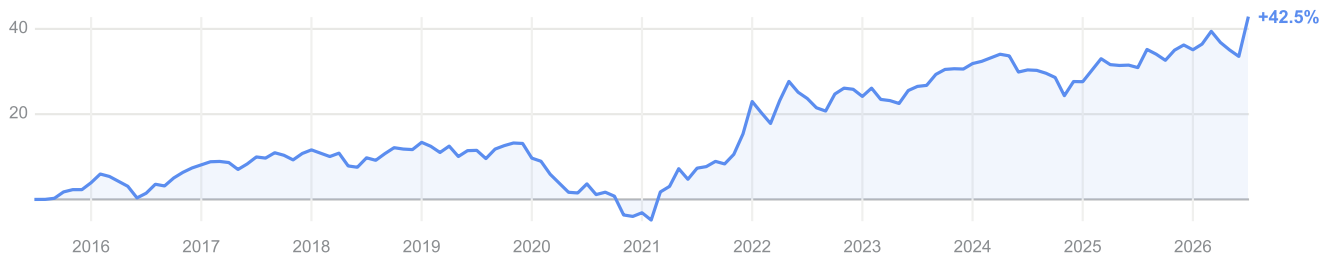
Capital Efficiency

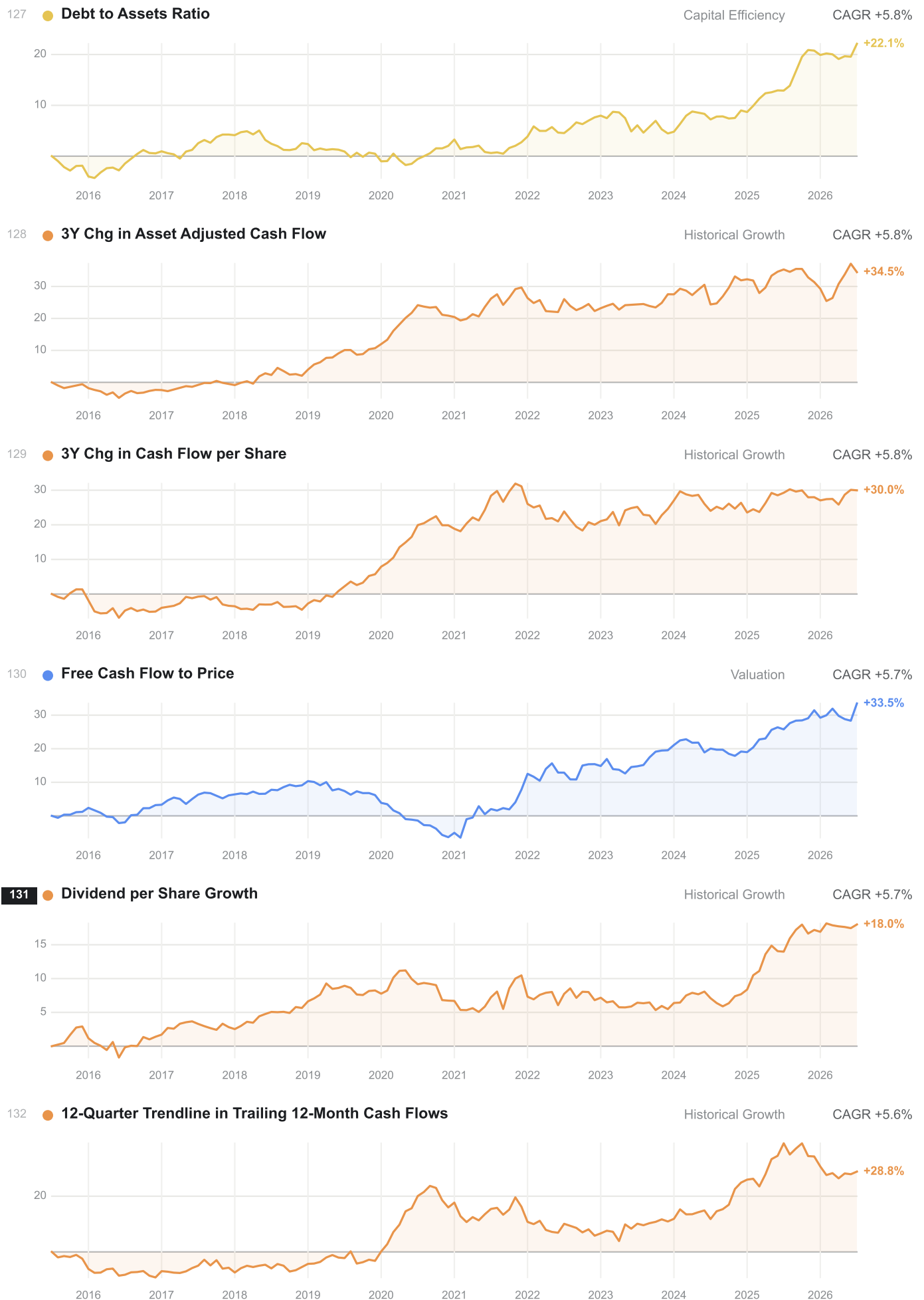
CAGR +5.8%

**126 ● Forward Free Cash Flow to Price Ratio**

Valuation

CAGR +5.8%

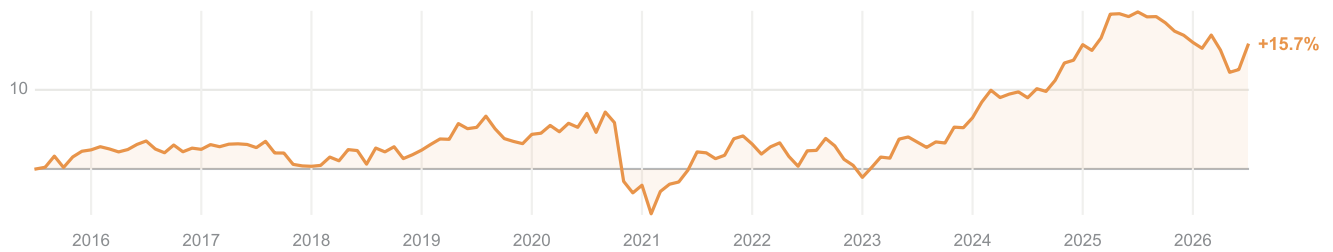




133 1Y Chg in Operating Cash Flow to Price

Historical Growth

CAGR +5.5%

**134 Cash Burn Rate**

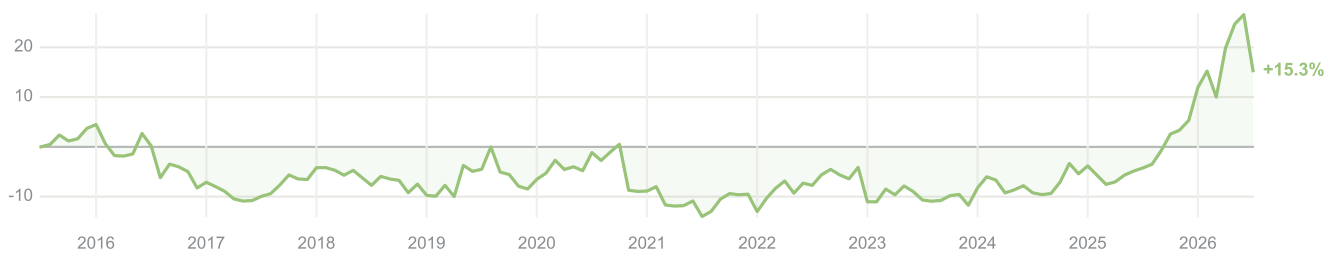
Earnings Quality

CAGR +5.5%

**135 52-Week Volume Price Trend with 20-Day Lag**

Price Momentum

CAGR +5.5%

**136 3Y Average Annual Earnings Growth**

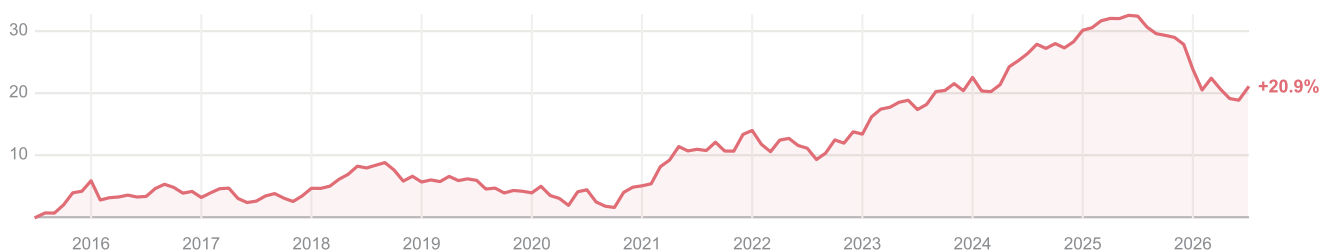
Historical Growth

CAGR +5.5%

**137 Depreciation to Capex Ratio**

Earnings Quality

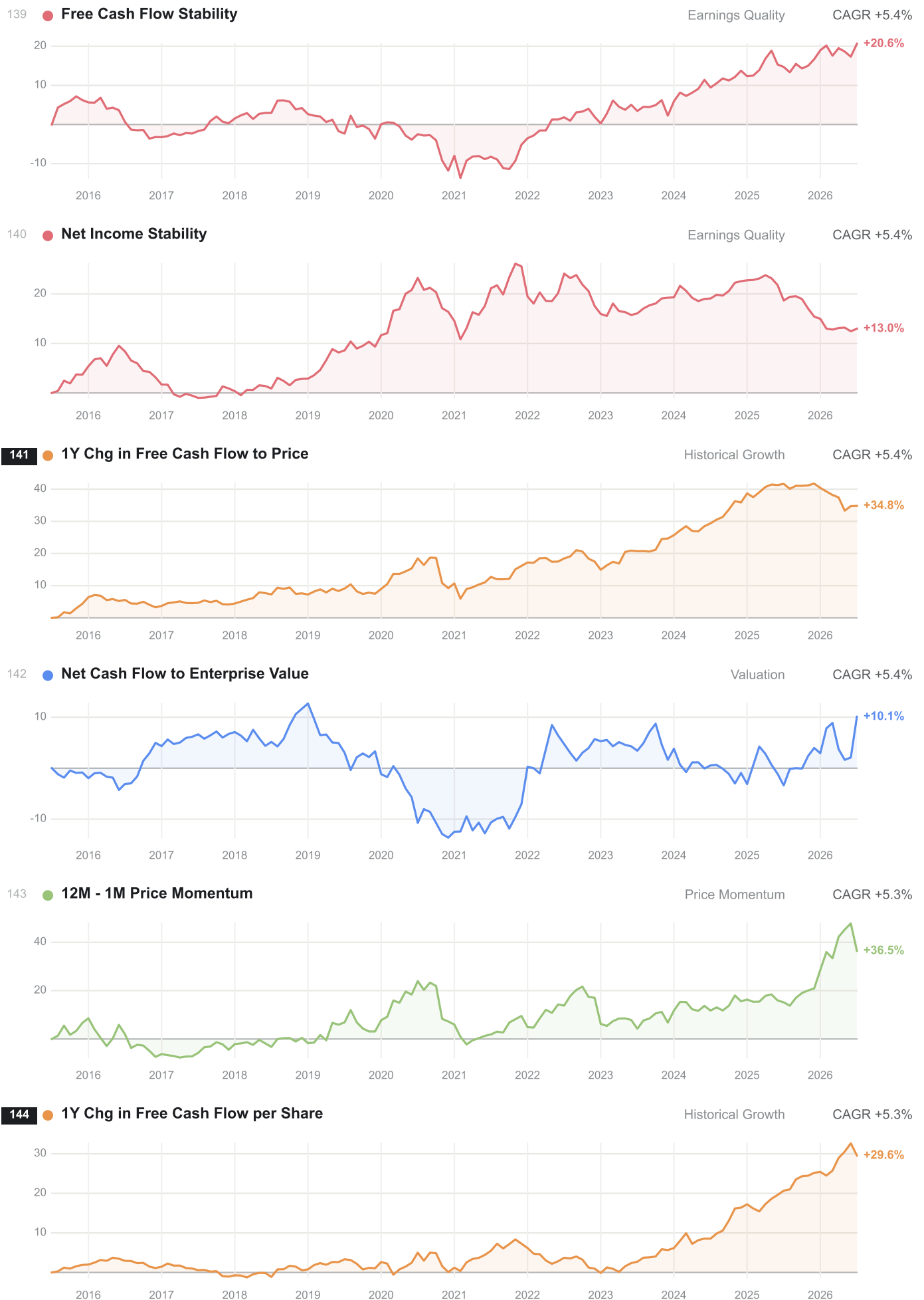
CAGR +5.5%

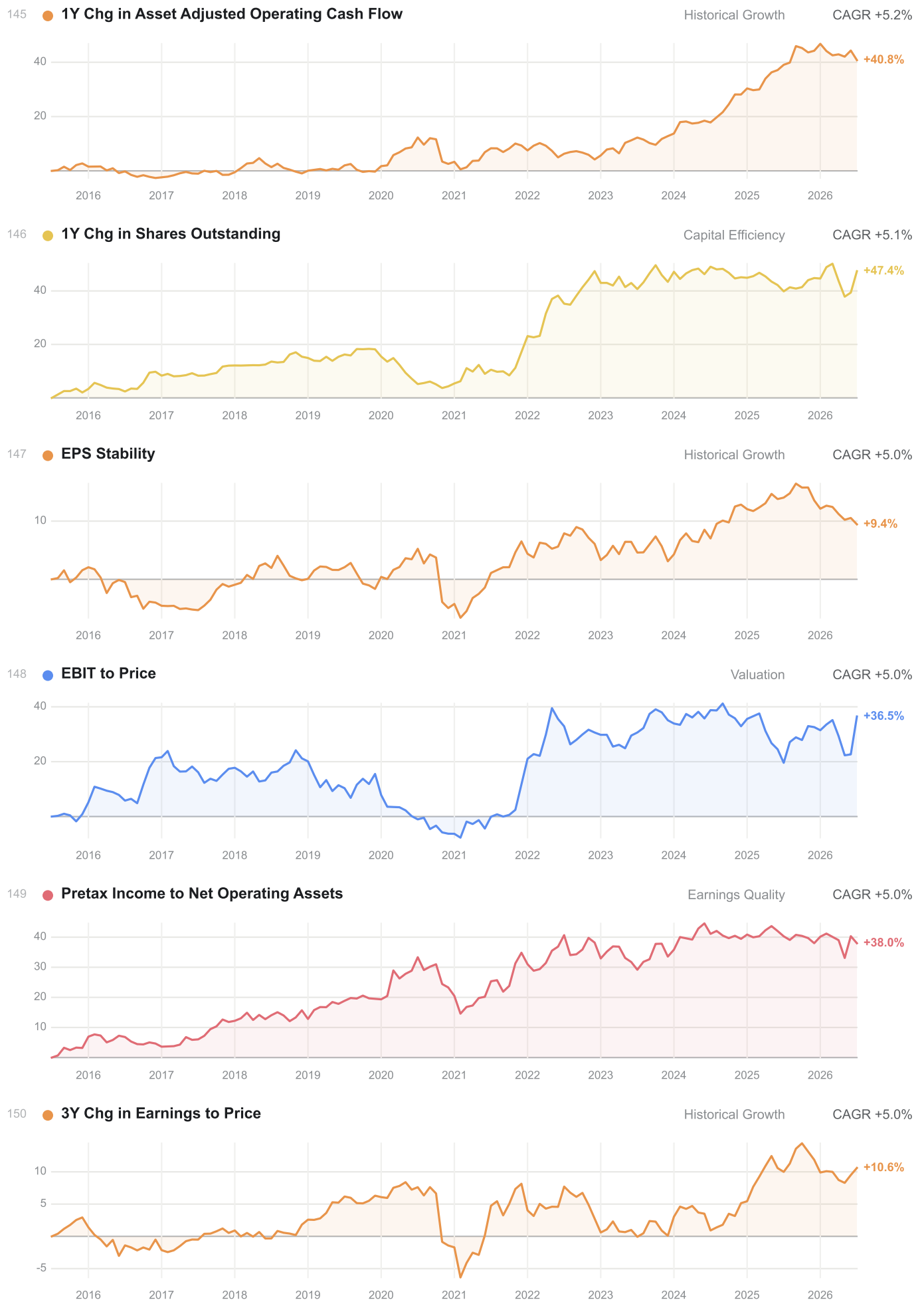
**138 Capital Acquisition Ratio**

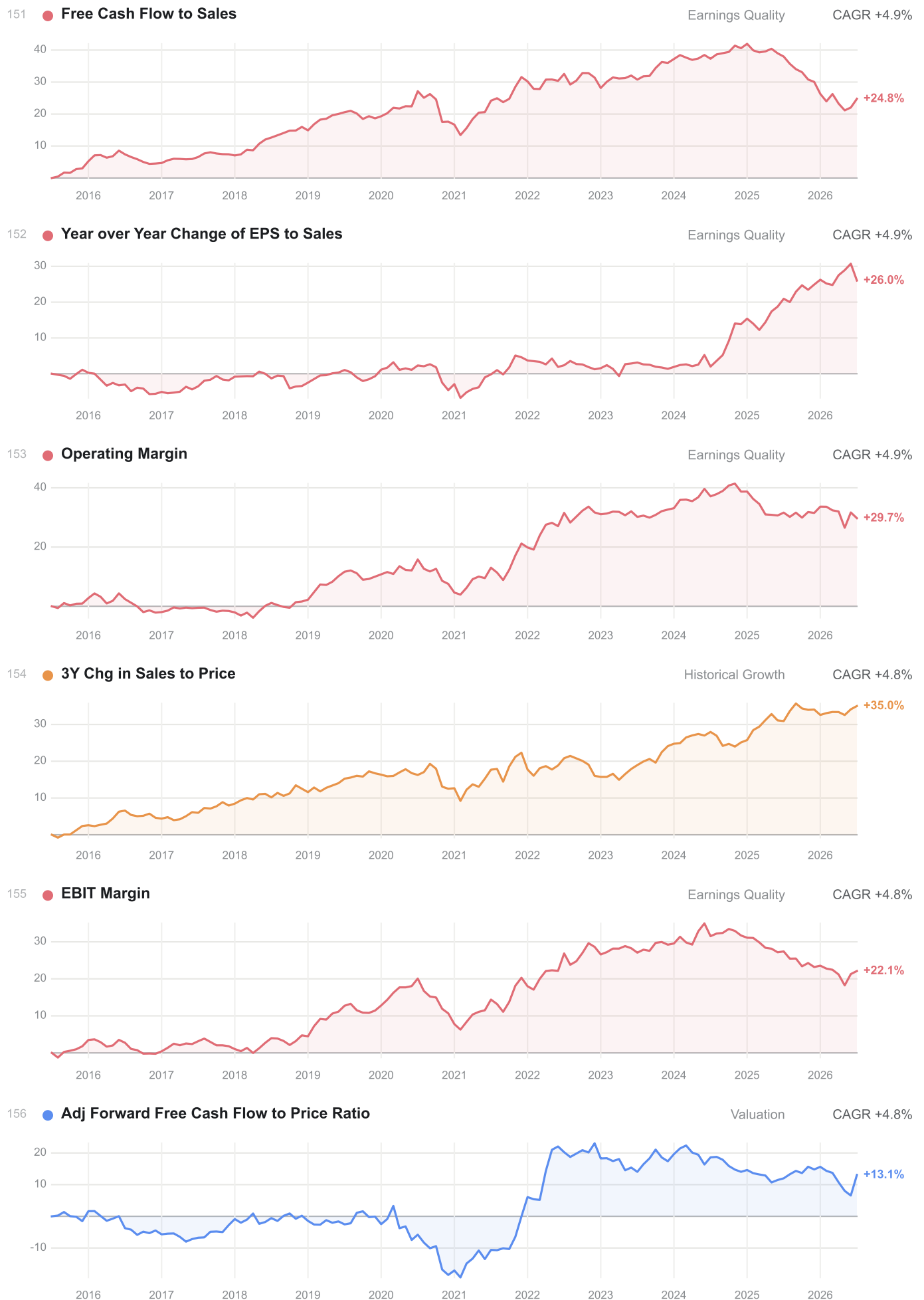
Capital Efficiency

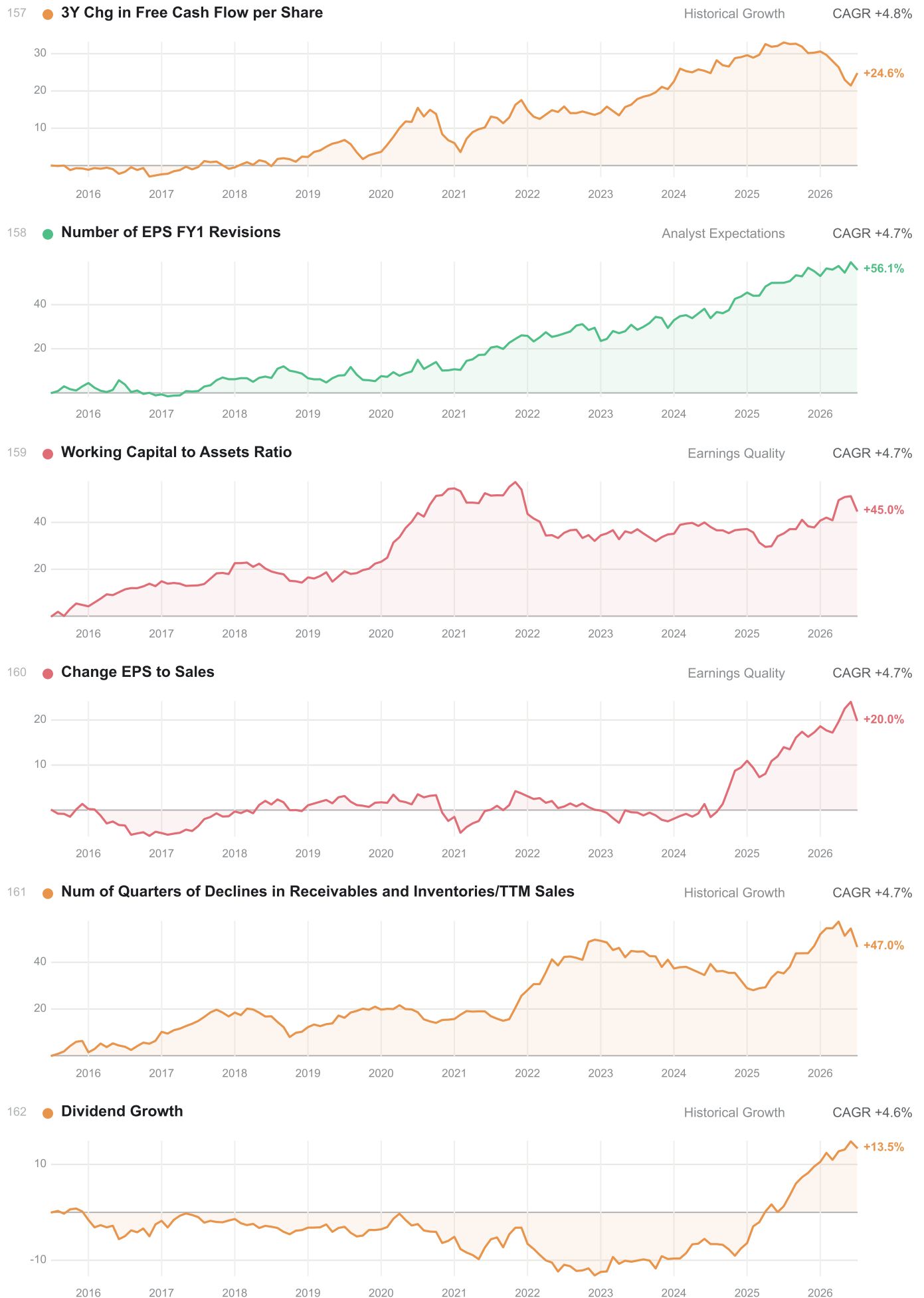
CAGR +5.5%

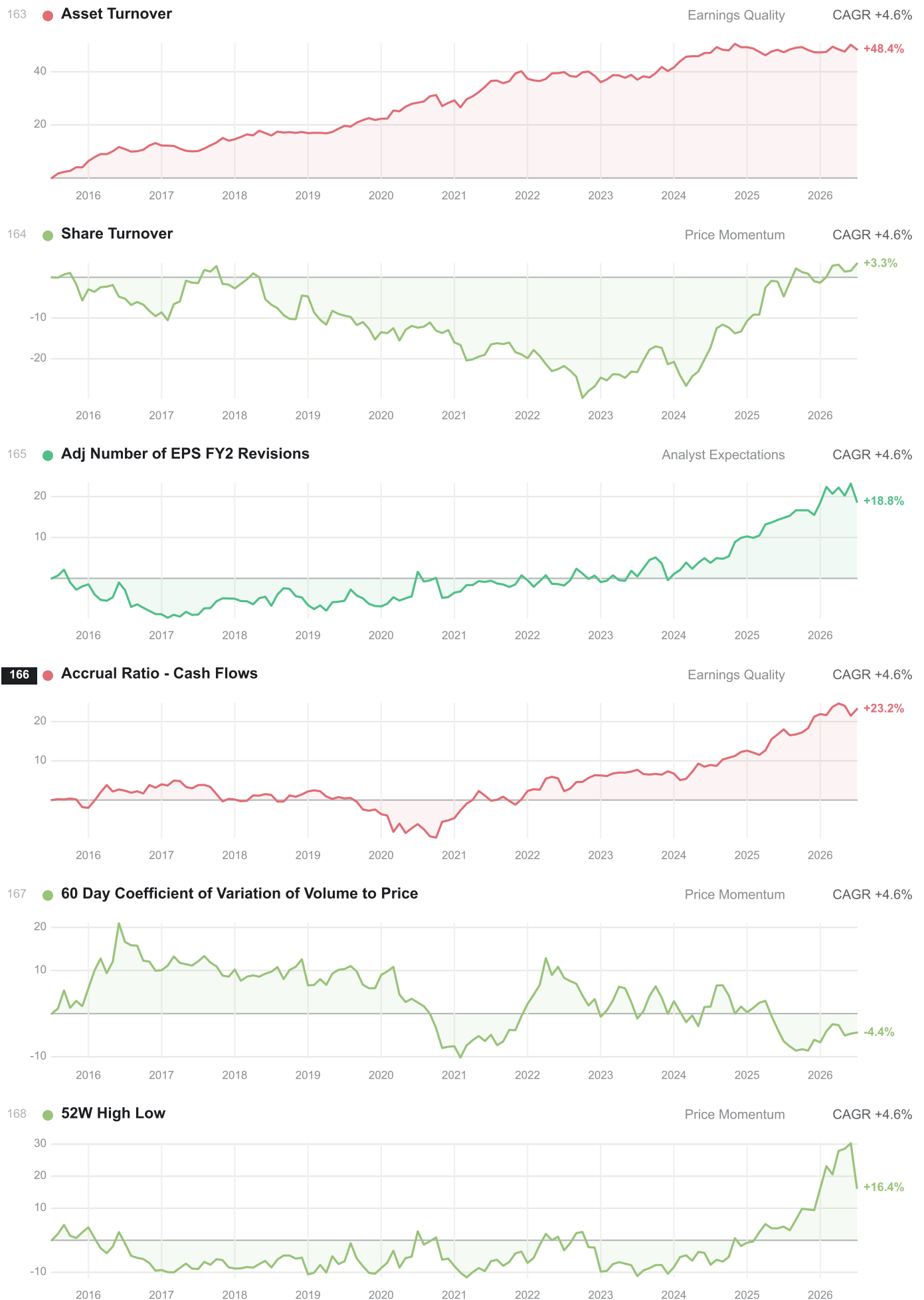


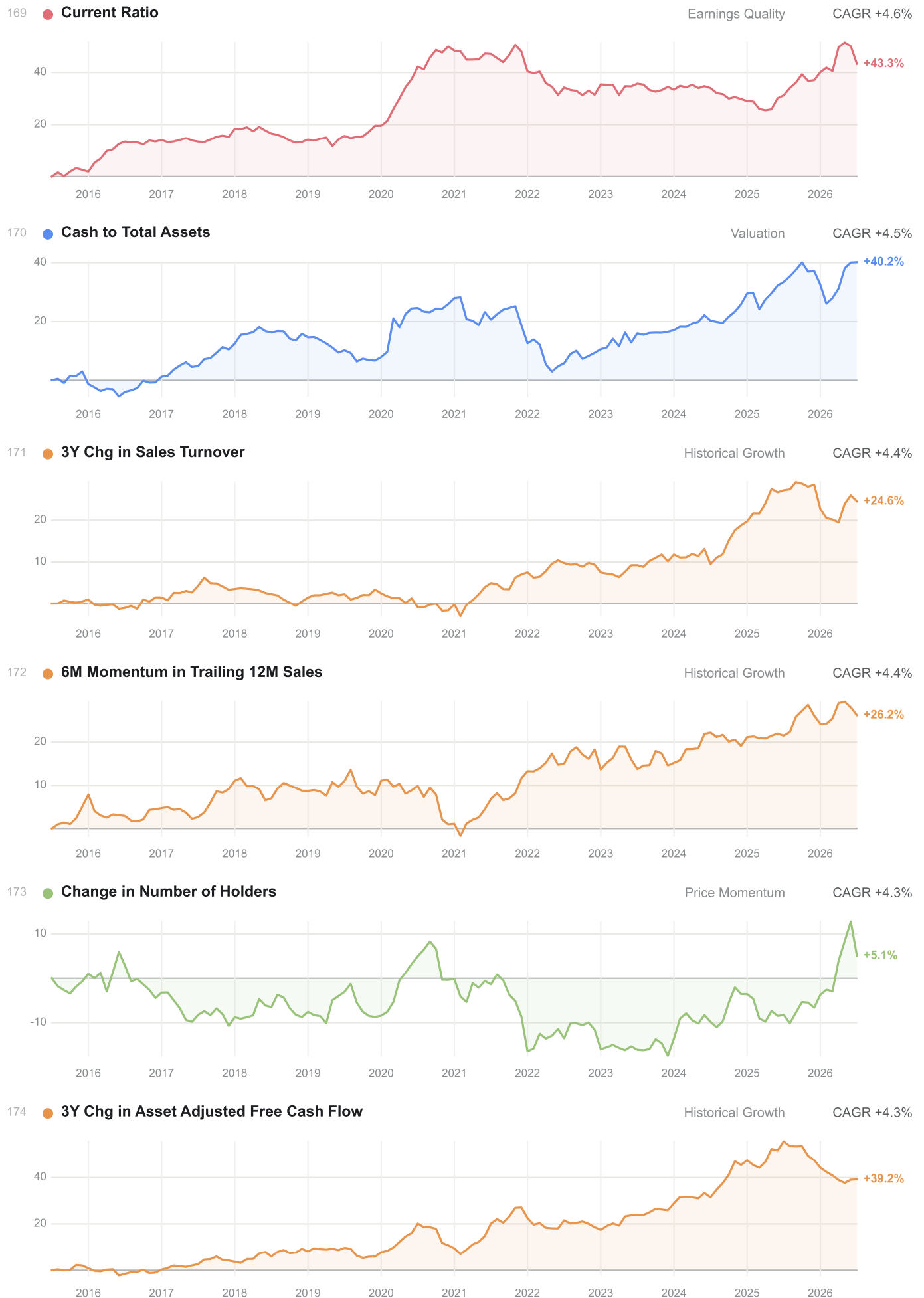


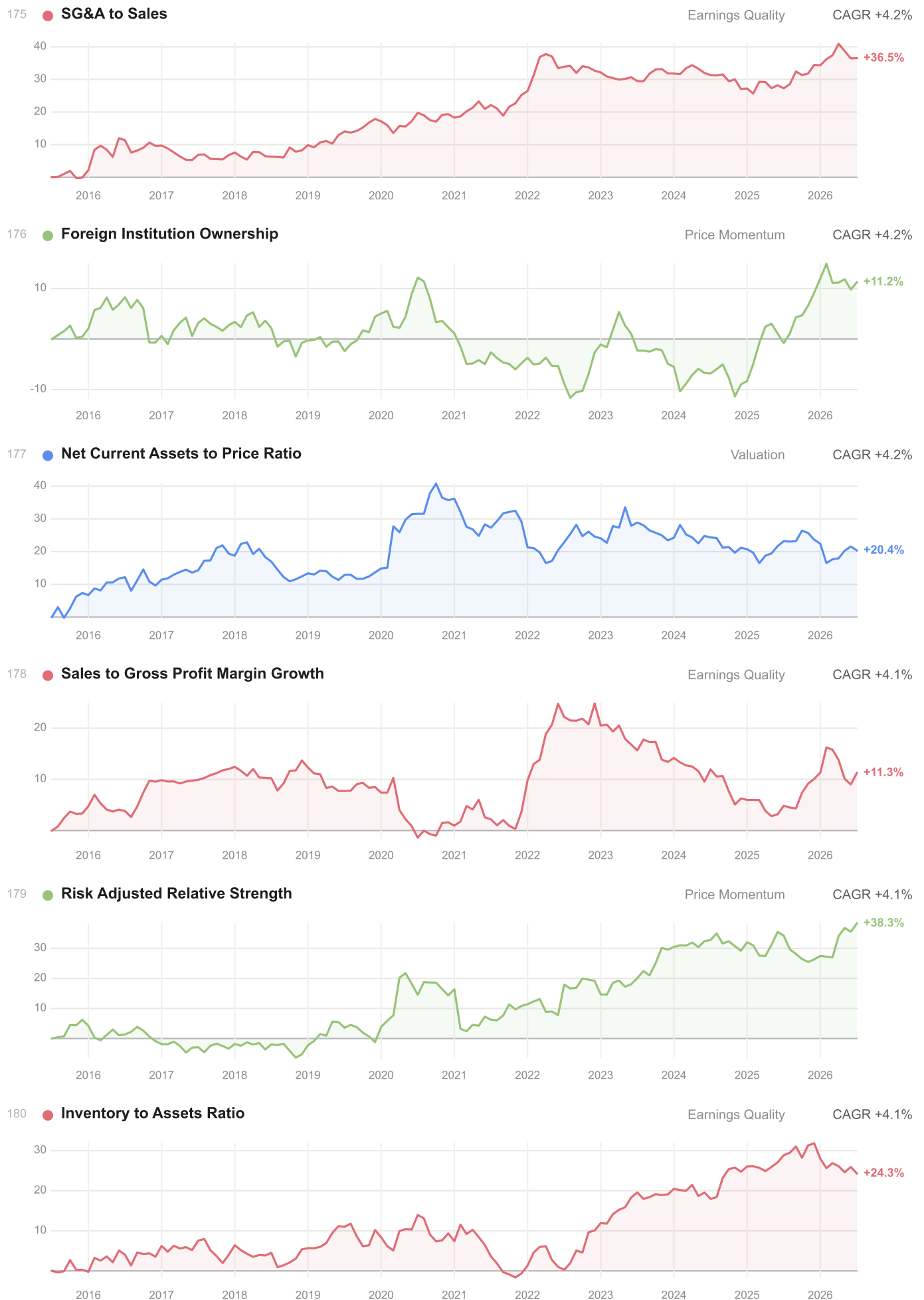


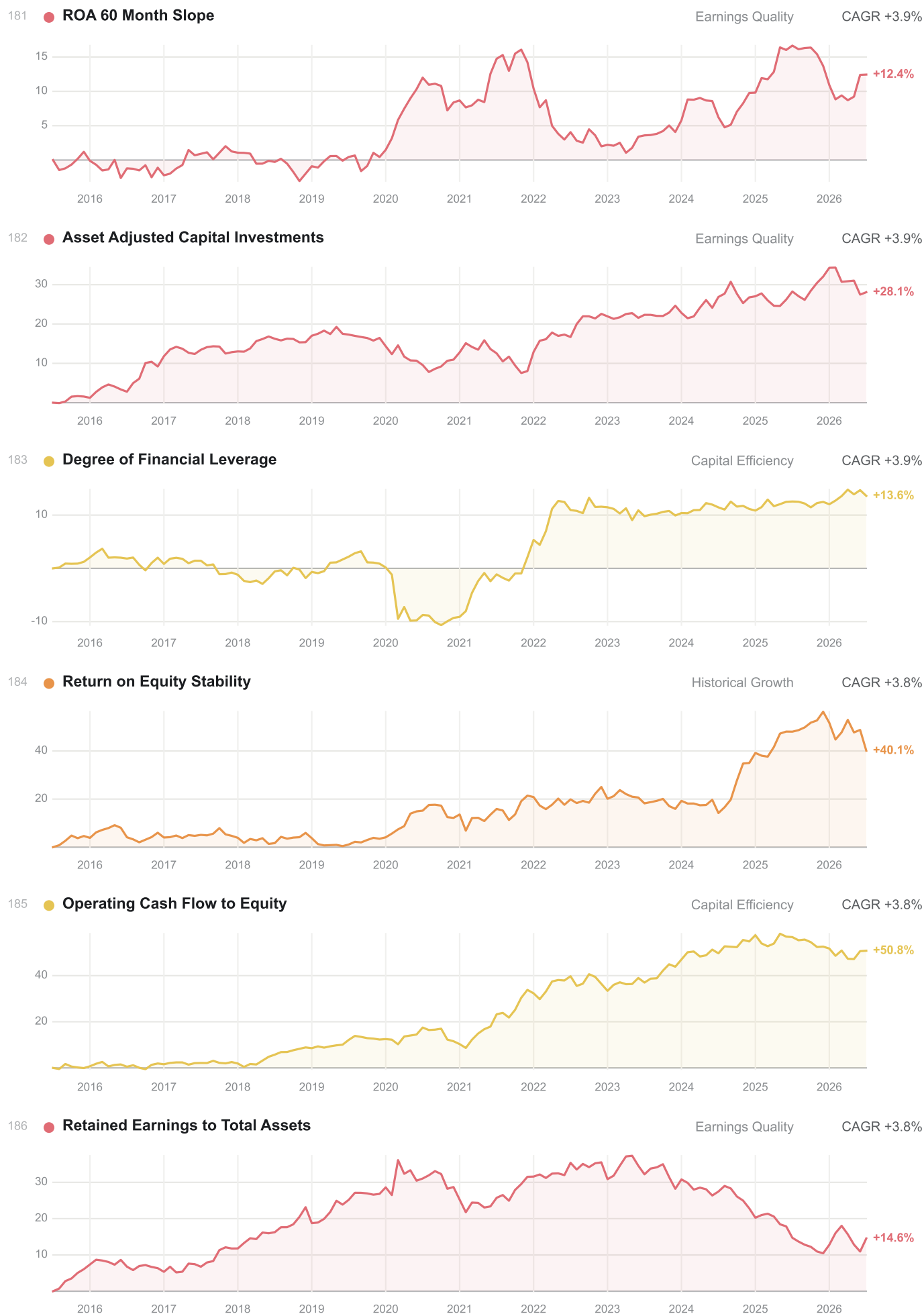


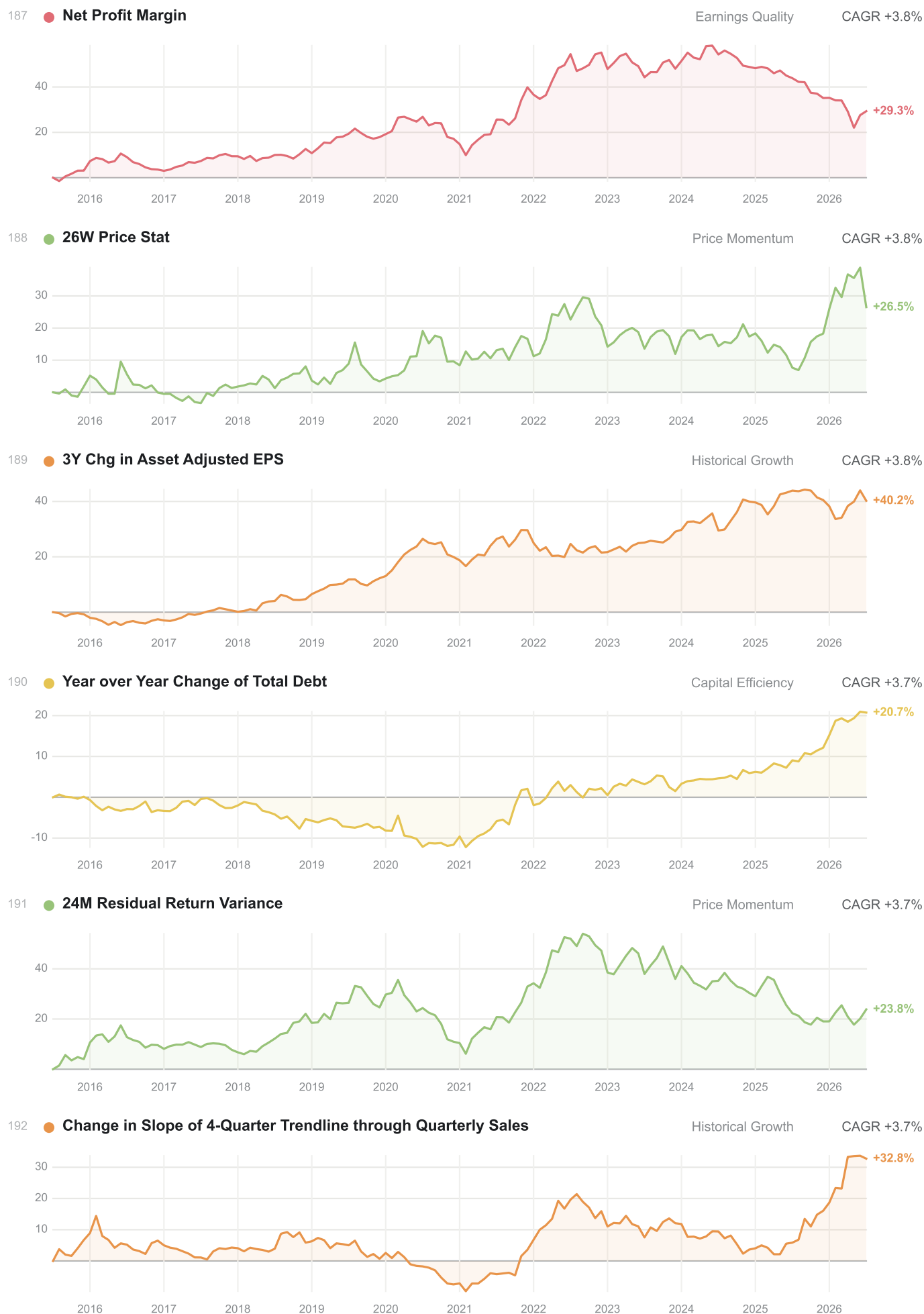


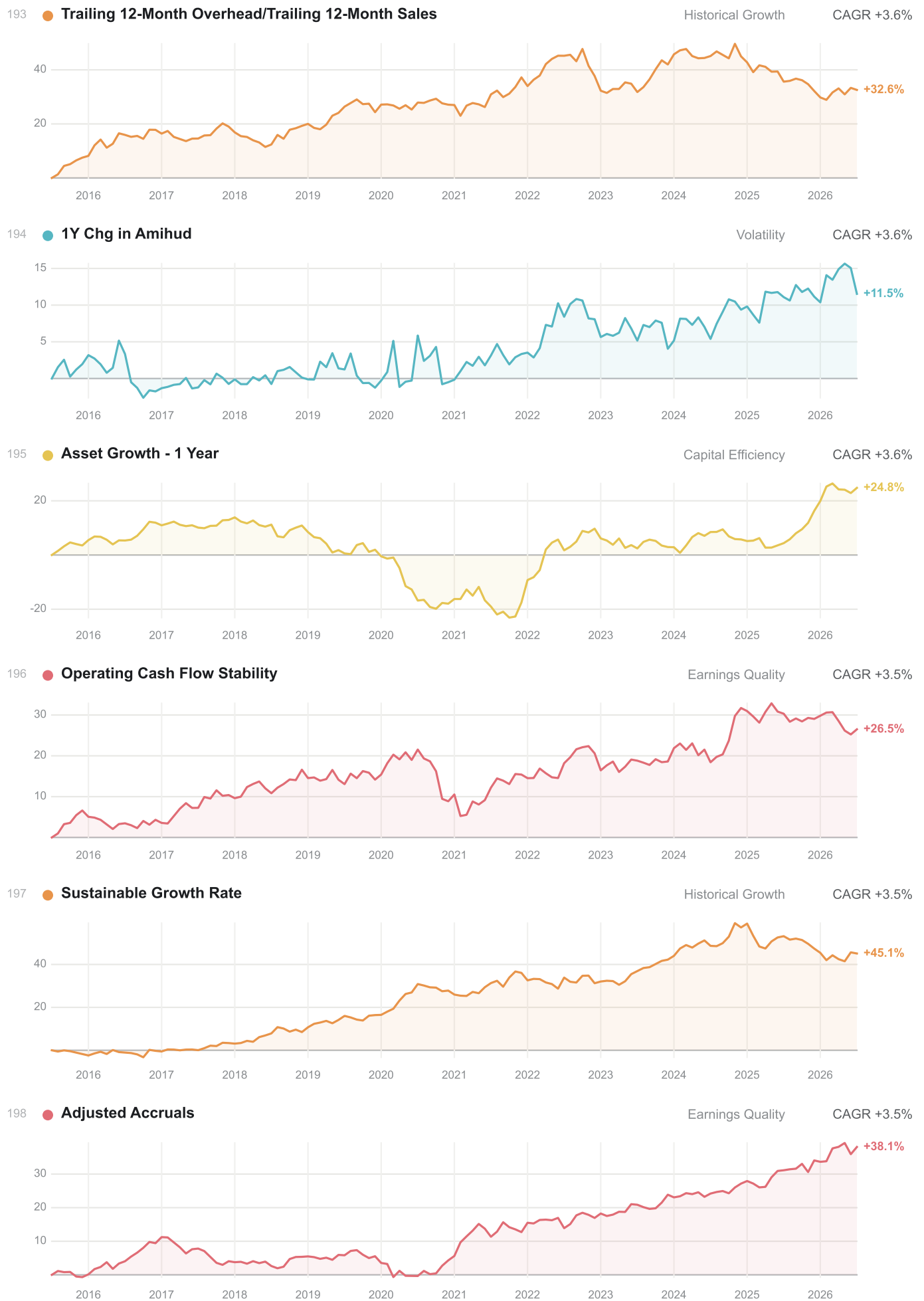


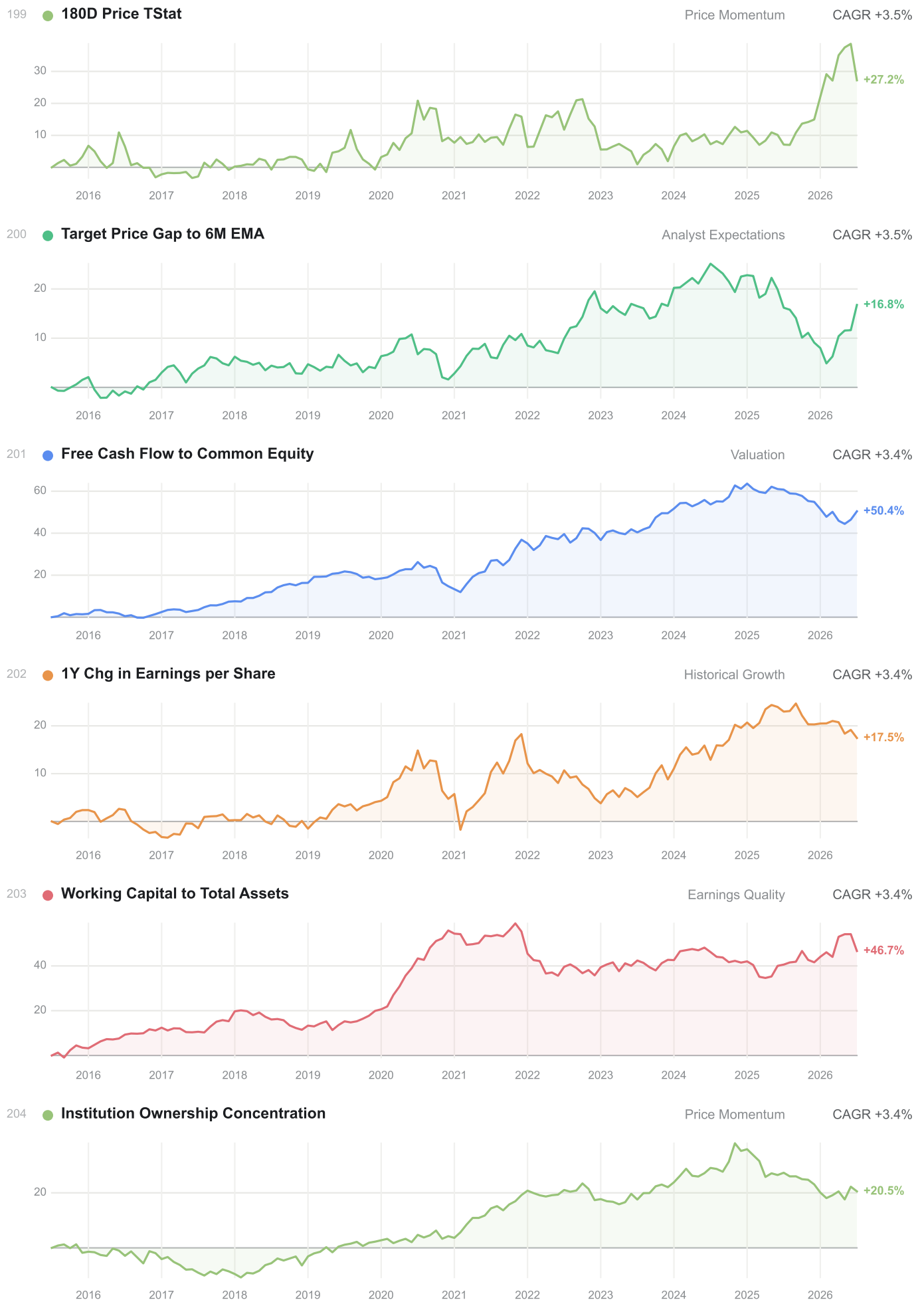


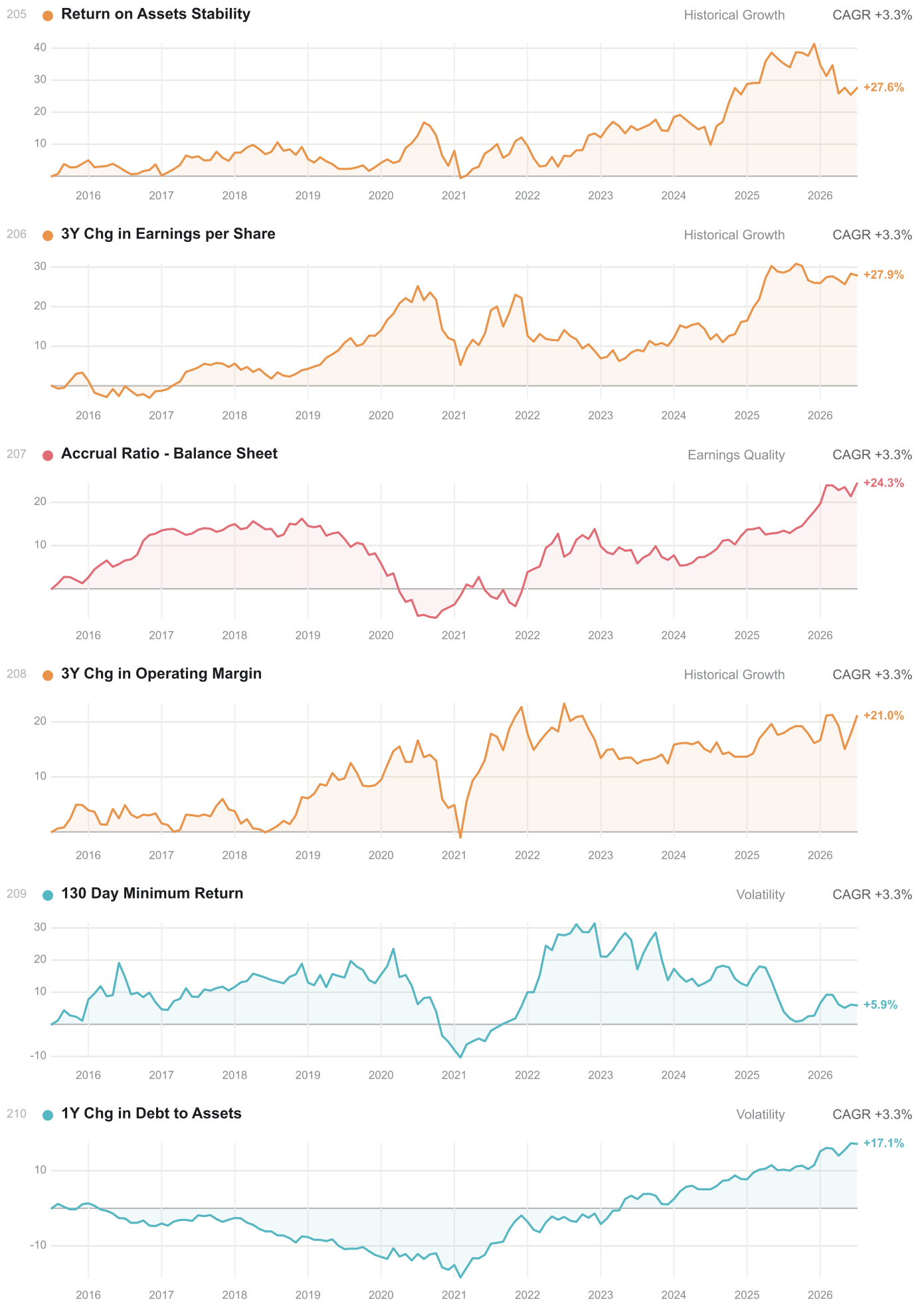


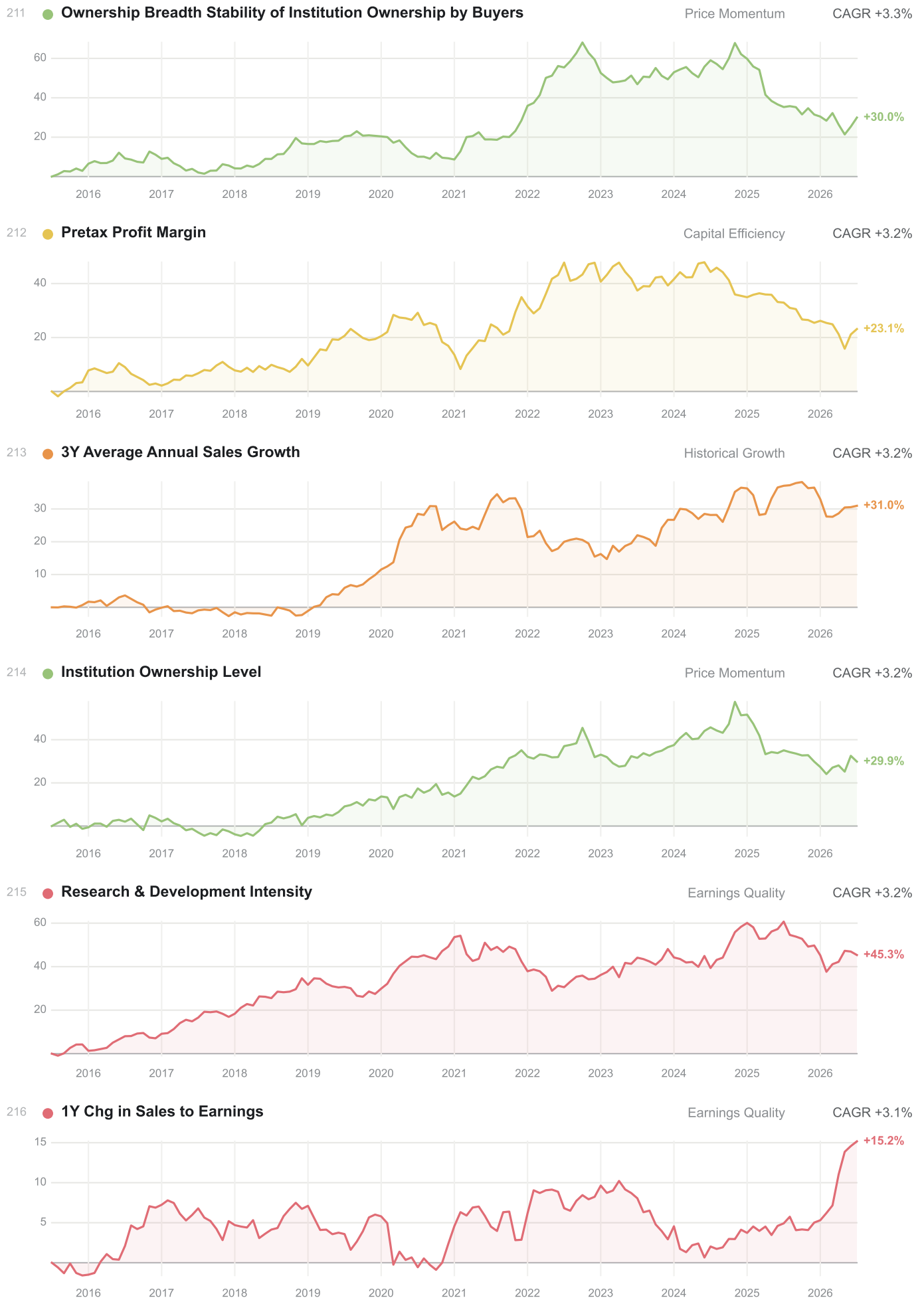


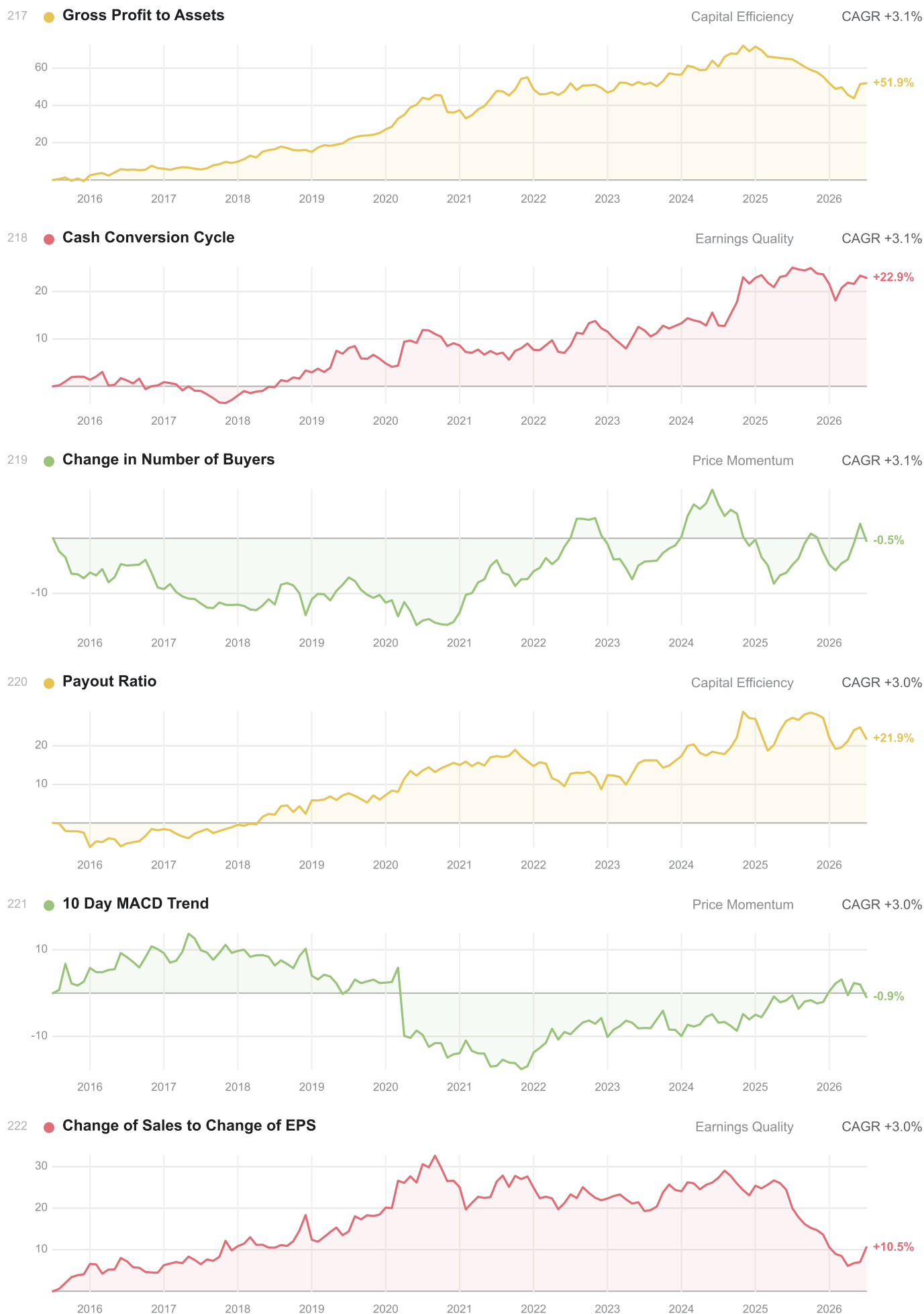












223 ● **6M Chg in 18M CAPM Alpha**

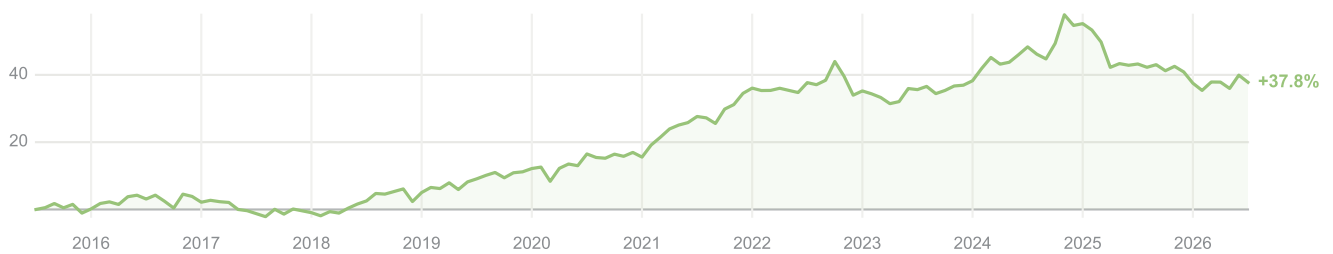
Price Momentum

CAGR +3.0%

224 ● **Investment Duration**

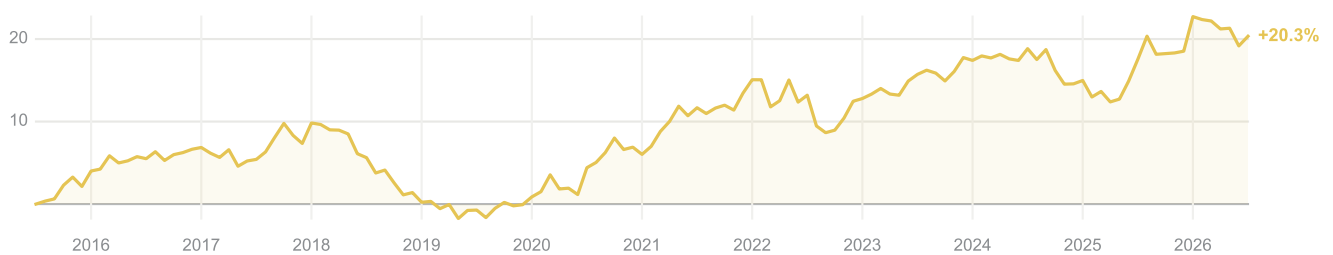
Price Momentum

CAGR +2.9%

225 ● **Working Capital Turnover Ratio**

Capital Efficiency

CAGR +2.9%

226 ● **Ownership Breadth Stability of Institution Ownership by Holders**

Price Momentum

CAGR +2.9%

227 ● **6M Chg in 36M CAPM Alpha**

Price Momentum

CAGR +2.9%

228 ● **3M Momentum in Trailing 12M Sales**

Historical Growth

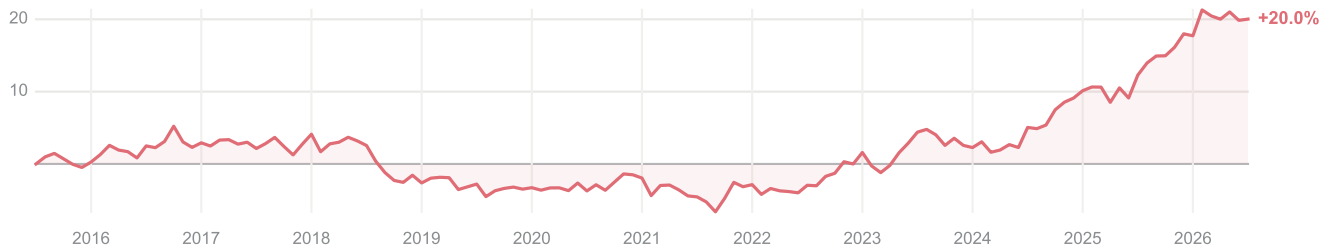
CAGR +2.9%



229 ● **Unexpected Receivables Change**

Earnings Quality

CAGR +2.8%

230 ● **Cash to Sales**

Earnings Quality

CAGR +2.8%

231 ● **EBIT Estimate Dispersion**

Analyst Expectations

CAGR +2.8%

232 ● **Adjusted Number of EPS FY1 Revisions**

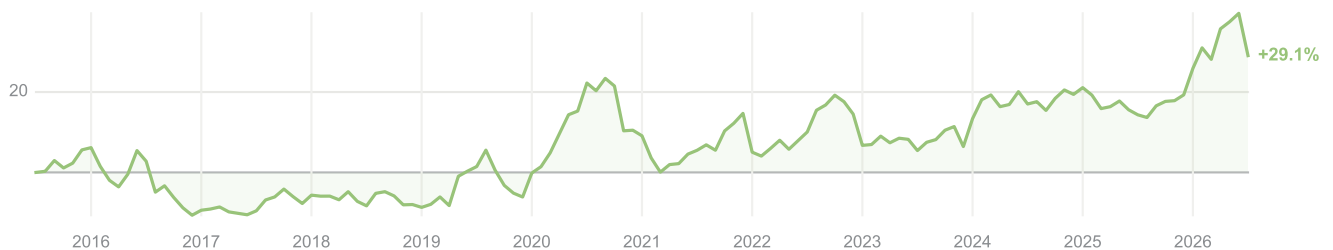
Analyst Expectations

CAGR +2.8%

233 ● **Volatility adjusted 12M return**

Price Momentum

CAGR +2.7%

234 ● **Standardized 36M Residual**

Price Momentum

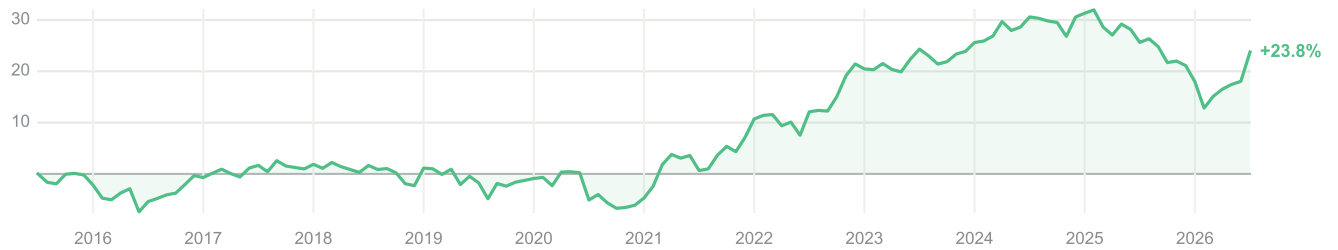
CAGR +2.6%



235 ● **6M Chg in Target Price Gap**

Analyst Expectations

CAGR +2.5%

236 ● **1Y Chg in EPS to Operating Cash Flow**

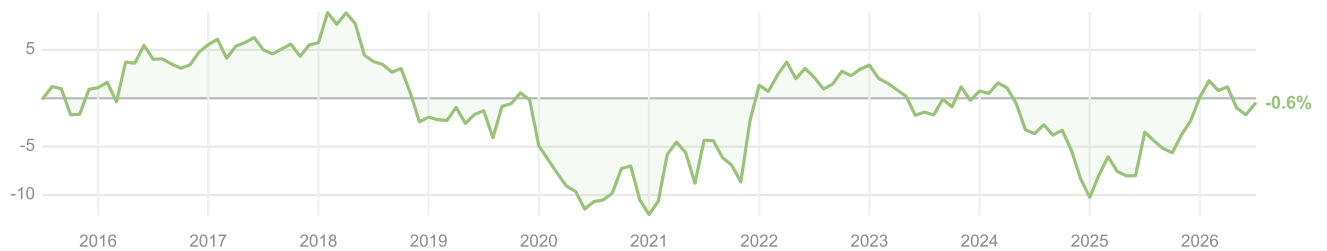
Earnings Quality

CAGR +2.4%

237 ● **Change of Ownership Level - Hedge Fund**

Price Momentum

CAGR +2.3%

238 ● **1Y Chg in Cash Flow per Share**

Historical Growth

CAGR +2.3%

239 ● **Adj 50 Day Volume Signal**

Volatility

CAGR +2.3%

240 ● **Analyst Earnings Estimate Diffusion**

Analyst Expectations

CAGR +2.2%



241 ● **Max Daily Return in the Past 6 Months**

Price Momentum

CAGR +2.1%

242 ● **6 Month Average Share Turn Over**

Price Momentum

CAGR +2.0%

243 ● **30 Day Coefficient of Variation of Volume to Price**

Price Momentum

CAGR +1.9%

244 ● **20 Day Coefficient of Variation of Volume to Price**

Price Momentum

CAGR +1.9%

245 ● **Year over Year Change of Inventory to Assets**

Earnings Quality

CAGR +1.8%

246 ● **Buy-to-Sell Recommendation Ratio Less 3M SMA**

Analyst Expectations

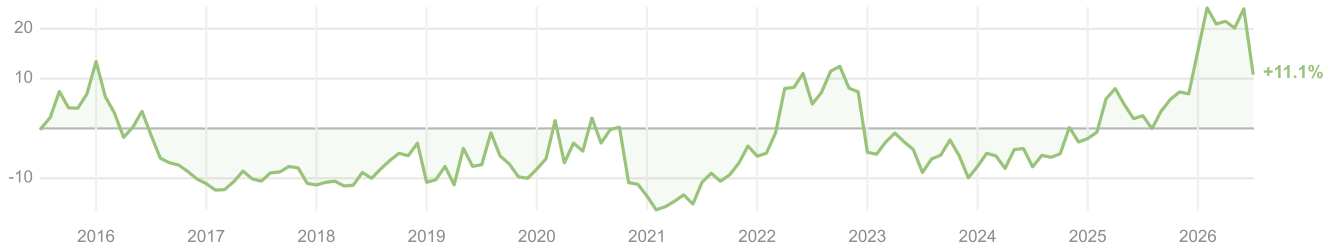
CAGR +1.8%



247 ● Closing Price to 52 Week High

Price Momentum

CAGR +1.8%



248 ● Institution Ownership Turnover

Price Momentum

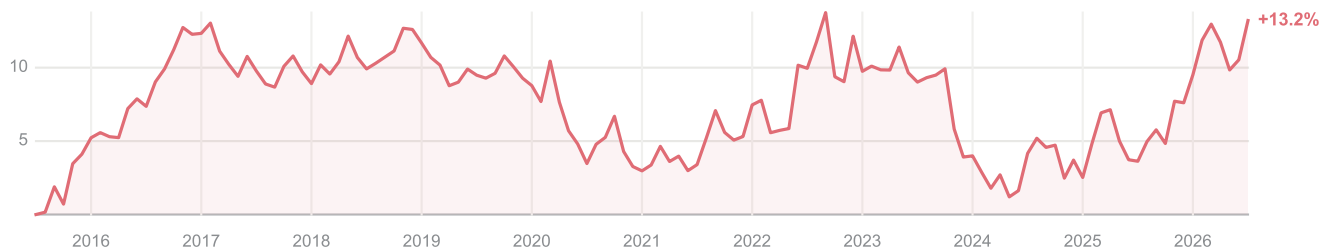
CAGR +1.7%



249 ● ROE 20 Qtr Standard Deviation

Earnings Quality

CAGR +1.5%



250 ● 6M Chg in 12M CAPM Alpha

Price Momentum

CAGR +1.3%



251 ● Sales Acceleration

Historical Growth

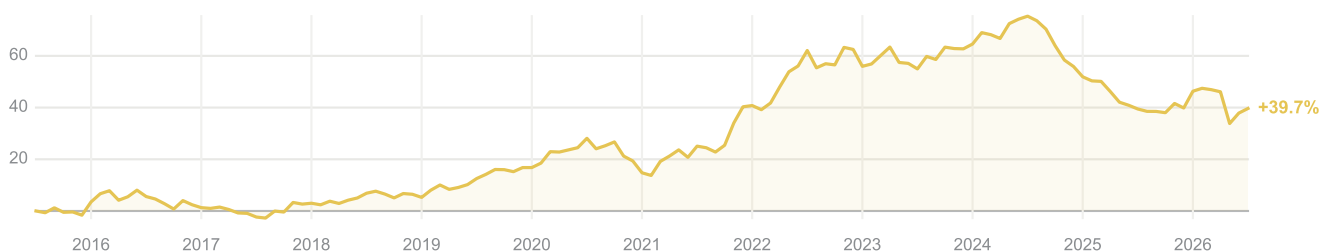
CAGR +1.3%



252 ● Normalized ROA

Capital Efficiency

CAGR +1.2%



253 ● **6M Avg Chg 1M Recommendation**

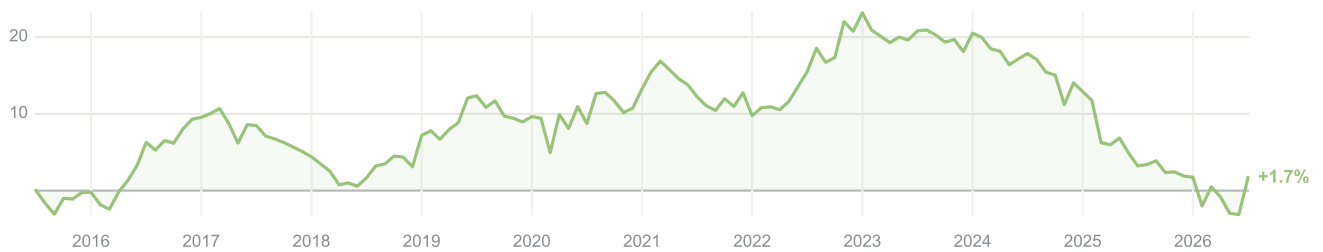
Analyst Expectations

CAGR +1.1%

254 ● **14-Day Relative Strength Index**

Price Momentum

CAGR +0.9%

255 ● **Change in TTM Depr. to Capex**

Earnings Quality

CAGR +0.8%

256 ● **Buy-to-Sell Recommendation Ratio Less 3M EMA**

Analyst Expectations

CAGR +0.6%

257 ● **Revenue Estimate Dispersion**

Analyst Expectations

CAGR +0.6%

258 ● **8-Week Change in 12-Month Forward Earnings Consensus Estimate/Price**

Valuation

CAGR +0.5%

